

universe and rendered it transparent. The independent evolution of radiation and matter began when the temperature had cooled sufficiently to allow the free electrons to combine with nuclei of hydrogen and helium. This formation of neutral atoms is sometimes referred to as **recombination**, an oddly inappropriate term since the electrons and nuclei had never been previously combined into atoms! The loss of free electrons and the resulting drop in opacity completed the decoupling of radiation and matter, freeing the photons to roam unhindered throughout a newly transparent universe. The photons of the cosmic microwave background that we observe today were last scattered during the time of recombination.

The Surface of Last Scattering

We define the **surface of last scattering** as a spherical surface, centered on the Earth, from which the CMB photons just now arriving at Earth were last scattered before beginning their unimpeded journey to us. (Of course, other observers in the universe are at the centers of their own surfaces of last scattering.) The surface of last scattering is the farthest redshift we can possibly observe at this moment in time. More accurately, because recombination did not happen all at once, the surface of last scattering has a thickness Δz . Just as the light from the Sun was last scattered from somewhere within its photosphere, the CMB photons originated within a layer, the “surface” of last scattering.²⁵ The surface of last scattering can therefore be thought of as a curtain that screens everything prior to decoupling from the direct view of astronomers. The earliest moments of the universe are hidden behind this veil and must be investigated indirectly.

The Conditions at Recombination

The temperature at recombination can be estimated through use of the Saha equation (Eq. 8.8),

$$\frac{N_{\text{II}}}{N_{\text{I}}} = \frac{2Z_{\text{II}}}{n_e Z_{\text{I}}} \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi_{\text{I}}/kT}.$$

Assuming (incorrectly) a composition of pure hydrogen for simplicity, we use $Z_{\text{I}} = 2$ and $Z_{\text{II}} = 1$ as in Example 8.1.4. It is useful to define f to be the fraction of hydrogen atoms that are ionized, so

$$f = \frac{N_{\text{II}}}{N_{\text{I}} + N_{\text{II}}} = \frac{N_{\text{II}}/N_{\text{I}}}{1 + N_{\text{II}}/N_{\text{I}}}, \quad (29.97)$$

or

$$\frac{N_{\text{II}}}{N_{\text{I}}} = \frac{f}{1 - f}. \quad (29.98)$$

²⁵The difference between the Sun’s photosphere and the surface of last scattering is that the photosphere has a spatial thickness, while the surface of last scattering has a thickness in terms of redshift or, equivalently, time.

For ionized hydrogen there is one free electron for every proton, $n_e = n_p$, so the number density of free electrons depends on f as

$$n_e = n_p = f(n_p + n_H) = \frac{f\rho_b}{m_H}, \quad (29.99)$$

where ρ_b is the density of baryonic matter. Note that in obtaining Eq. (29.99) from Eq. (29.97), N_I corresponds to n_H , the number density of neutral hydrogen atoms, and N_{II} corresponds to n_p , the number density of protons (ionized hydrogen atoms). Using Eq. (29.82), we can write this as

$$n_e(R) = \frac{f\rho_{b,0}}{m_H R^3}, \quad (29.100)$$

Substituting Eqs. (29.98) and (29.100) into the Saha equation, together with Eq. (29.58) for the blackbody temperature, we find

$$\frac{f}{1-f} = \frac{m_H R^3}{f\rho_{b,0}} \left(\frac{2\pi m_e k T_0}{h^2 R} \right)^{3/2} e^{-\chi_I R/kT_0}, \quad (29.101)$$

where $T_0 = 2.725$ K and $\chi_I = 13.6$ eV. This can be solved numerically to find that the universe had cooled sufficiently for one-half of its electrons and protons to combine to form atomic hydrogen ($f = 0.5$) when the value of the scale factor was approximately $R \approx 7.25 \times 10^{-4}$ ($z \approx 1380$), corresponding to a temperature of about 3760 K [again from Eq. (29.58)].

More precisely, the WMAP value for the redshift at the time of decoupling (i.e., the surface of last scattering) is

$$[z_{\text{dec}}]_{\text{WMAP}} = 1089 \pm 1.$$

We will adopt this as the value of the redshift for both recombination and decoupling. Using Eqs. (29.4) and (29.58) yields a temperature at recombination of

$$T_{\text{dec}} = T_0(1 + z_{\text{dec}}) = 2970 \text{ K}.$$

This is lower than our estimate of 3760 K because the photons created by the formation of some atoms were then absorbed by other atoms, putting these atoms into excited states from which they were easier to ionize. Thus a slightly cooler temperature was needed to complete the recombination process. It is important to remember that at times earlier than recombination, the radiation and matter shared a common temperature, whereas after recombination, the temperatures of the radiation and matter must be distinguished. For the remainder of the text, it is the radiation temperature (the temperature of the CMB) that will be of interest after recombination.

The WMAP value for the time at which recombination and decoupling occurred is

$$[t_{\text{dec}}]_{\text{WMAP}} = 379_{-7}^{+8} \text{ kyr}. \quad (29.102)$$

Of course, these events did not occur at a single instant of time; the WMAP value of the decoupling time interval is

$$[\Delta t_{\text{dec}}]_{\text{WMAP}} = 118_{-2}^{+3} \text{ kyr.}$$

This corresponds to the surface of last scattering having a thickness (in redshift) of

$$[\Delta z_{\text{dec}}]_{\text{WMAP}} = 195 \pm 2.$$

The Dawn of Precision Cosmology

Before ending this section, it is worth taking a few moments to reflect on the power of simple mathematics to illuminate the earliest stages of the universe. Before the discovery of the cosmic microwave background, theorizing about the early universe was not considered to be legitimate science by many physicists. The point is well made by Steven Weinberg in his book *The First Three Minutes*:

This is often the way it is in physics—our mistake is not that we take our theories too seriously, but that we do not take them seriously enough. It is always hard to realize that these numbers and equations we play with at our desks have something to do with the real world. . . . The most important thing accomplished by the ultimate discovery of the 3° K radiation background in 1965 was to force us all to take seriously the idea that there *was* an early universe.

The results of the Wilkinson Microwave Anisotropy Probe have now taken cosmology to a level that was hardly imaginable when the first edition of *An Introduction to Modern Astrophysics* was published in 1996. In remarks made at the WMAP press conference on February 11, 2003, astrophysicist John Bahcall said,

Every astronomer will remember where he or she was when they first heard the WMAP results. I certainly will. . . . For cosmology, the formal announcement today represents a “rite of passage” from speculation to precision science.

29.3 ■ RELATIVISTIC COSMOLOGY

The appearance of objects at truly cosmological distances is affected by the curvature of the spacetime through which the light travels on its way to Earth. The geometrical properties of the universe as a whole will be best understood by starting with simple analogies before proceeding to the more comprehensive description that only Einstein’s general theory of relativity can provide.

Euclidean, Elliptic, and Hyperbolic Geometries

The foundations of plane geometry (appropriate for a flat universe) were laid by Euclid sometime around 300 B.C. Euclid’s *Elements* consists of 13 “books” (chapters) that contain 465 theorems. These theorems, in turn, are derived from just five postulates that embody self-evident truths and so are stated without proof. In Euclid’s words,

1. It is possible to draw a straight line from any point to any point.
2. It is possible to produce [extend] a finite straight line continuously in a straight line.
3. It is possible to describe a circle with any center and distance [radius].
4. All right angles are equal to one another.

Euclid's statement of his fifth postulate is awkward and opaque:

5. If a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on the side on which the angles are less than the two right angles.

A moment's thought will show that Euclid's fifth postulate concerns the behavior of parallel lines.²⁶ It is equivalent to the following statement, due to an English mathematician, John Playfair (1748–1819):

- 5a. Given, in a plane, a line L and a point P not on L , then through P there exists one and only one line parallel to L .

Figure 29.15(a) illustrates Playfair's version of the parallel postulate in Euclidean geometry.

The cumbersome statement of Euclid's fifth postulate led some mathematicians to suspect that it might be derived from the first four. In the nineteenth century, mathematicians mounted an attack on the parallel postulate by denying it, in hopes that they would reach a contradiction. Such a proof by contradiction would make the fifth postulate dependent on the rest of geometry and so abolish its stature as an independent postulate. To their astonishment, instead of arriving at a contradiction as they expected, several renowned mathematicians realized that they had developed *alternative* geometries, each as self-consistent as Euclid's.

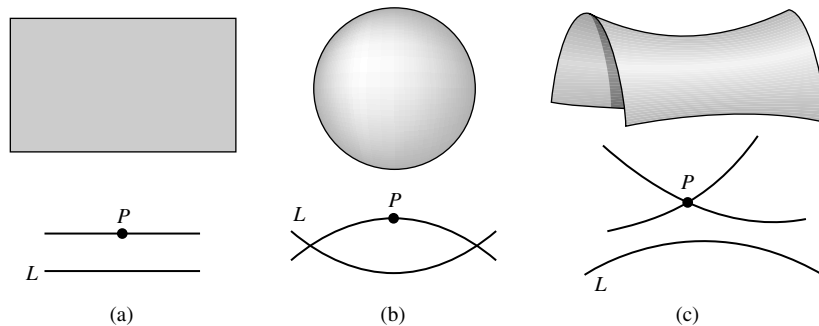


FIGURE 29.15 The parallel postulate, illustrated for three alternative geometries: (a) Euclidean, (b) elliptic, and (c) hyperbolic.

²⁶Euclid defines parallel lines as “straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.”

Foremost among these mathematicians were Carl Frederick Gauss (1777–1855), Nikolai Lobachevski (1793–1856), János Bolyai (1802–1860), and Bernhard Riemann (1826–1866). There are *three* different fifth postulates, each of which leads to a perfectly well-behaved geometry. In 1868 it was proved that the two additional, non-Euclidean geometries are as logically consistent as Euclid’s original version. In addition to Playfair’s version of the parallel postulate given above for Euclidean geometry, there is the fifth postulate of the elliptic geometry developed by Riemann:

- 5b. Given, in a plane, a line L and a point P not on L , then through P there exists *no* line parallel to L .

As Fig. 29.15(b) shows, this describes the geometry of the surface of a sphere, where two lines that both start out perpendicular to the sphere’s equator meet at its poles. In elliptic geometry, the angles of a triangle add up to *more* than 180° , and the circumference of a circle is less than $2\pi r$.

On the other hand, the fifth postulate of the hyperbolic geometry, developed independently by Gauss, Bolyai, and Lobachevski, is more generous than Euclid’s:

- 5c. Given, in a plane, a line L and a point P not on L , then through P there exist *at least two* lines parallel to L .

Figure 29.15(c) shows this geometry applied to a saddle-shaped hyperboloid. Neither of the two lines shown passing through point P intersects the line L , and it is possible to draw (in this example) infinitely many more. In hyperbolic geometry, the angles of a triangle add up to *less* than 180° , and the circumference of a circle exceeds $2\pi r$.

The logical independence of the parallel postulate means that it cannot be derived from Euclid’s earlier postulates. Which geometry to adopt is an arbitrary choice, since all three are equally valid from a mathematical viewpoint. Which of these three geometries describes the spatial structure of the physical universe is a question that must be answered *empirically*, by observation. Gauss himself conducted such an experiment beginning in 1820, when he directed a survey of the Germanic state of Hanover. After carefully measuring the distances between three mountaintops that formed a huge triangle (the longest side was 107 km), he determined that the sum of the triangle’s angles was 180° —within experimental error. Gauss’s experiment was not sensitive enough to measure the curvature of space near Earth’s surface.

The Robertson–Walker Metric for Curved Spacetime

The spacetime surrounding any massive object is indeed curved. As described in Section 17.2, the spatial curvature is revealed in the radial term of the Schwarzschild metric, Eq. (17.22). However, the Schwarzschild metric is valid only outside matter. We will have to find another metric to describe the spacetime of the dust-filled universe of Section 29.1. Although a derivation of this metric is beyond the scope of this book, the following argument should help elucidate some of its properties. Our search for the metric is made somewhat easier by the cosmological principle. In a homogeneous and isotropic universe, although the curvature of space may change with time, the curvature must have the same value everywhere at a given time since the Big Bang.

Let's begin by considering the curvature of a two-dimensional surface. "Curvature" has a precise mathematical meaning; for example, a sphere of radius R has a constant curvature, K , that is defined to be $K \equiv 1/R^2$. Gauss realized that the curvature of any surface can be determined *locally*, on an arbitrarily small patch of its surface. It is instructive to imagine how the curvature of a sphere of radius R might be measured by a tiny (but talented) ant on the sphere's surface. It is free to roam about the sphere's surface but is unable to gain an outside perspective and view the sphere as a whole. How could the inquisitive ant determine the sphere's curvature?

Starting at the north pole of the sphere (call this point P ; see Fig. 29.16), the ant could mark a series of other points all of which are a distance D from P . When connected, these points form a circle with P at its center. The ant now measures the circumference, C_{meas} , of the circle and compares it with the expected value of $C_{\text{exp}} = 2\pi D$. However, the two values do not agree; since $D = R\theta$,

$$C_{\text{exp}} = 2\pi R\theta$$

while

$$C_{\text{meas}} = 2\pi R \sin \theta = 2\pi R \sin(D/R).$$

When the ant divides the fractional discrepancy between these two values of the circumference by the expected area of the circle, $A_{\text{exp}} = \pi D^2$, and then multiplies by 6π , the result (in the limit $D \rightarrow 0$) is the curvature of the sphere. That is, the ant starts with

$$6\pi \cdot \frac{C_{\text{exp}} - C_{\text{meas}}}{C_{\text{exp}} A_{\text{exp}}} = 6\pi \cdot \frac{2\pi D - C_{\text{meas}}}{(2\pi D)(\pi D^2)} = \frac{3}{\pi} \frac{2\pi D - C_{\text{meas}}}{D^3}.$$

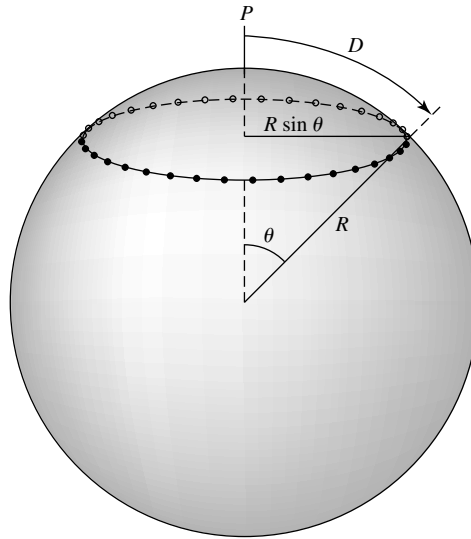


FIGURE 29.16 A local measurement of the curvature of a sphere.

Substituting for C_{meas} , the clever ant then uses a Taylor series for $\sin(D/R)$,

$$\begin{aligned} \frac{3}{\pi} \frac{2\pi D - 2\pi R \sin(D/R)}{D^3} &= \frac{6}{D^3} \left\{ D - R \left[\frac{D}{R} - \frac{1}{3!} \left(\frac{D}{R} \right)^3 + \frac{1}{5!} \left(\frac{D}{R} \right)^5 + \cdots \right] \right\} \\ &= \frac{1}{R^2} - \frac{1}{20} \frac{D^2}{R^4} + \cdots \end{aligned}$$

In the limit $D \rightarrow 0$, this is $1/R^2$, the curvature of the sphere. In fact, the prescription

$$K = \frac{3}{\pi} \lim_{D \rightarrow 0} \frac{2\pi D - C_{\text{meas}}}{D^3} \quad (29.103)$$

can be used to calculate the curvature at any point on a two-dimensional surface; see Fig. 29.17. For a flat plane, $K = 0$, while for a saddle-shaped hyperboloid, K is *negative* because the measured circumference exceeds $2\pi D$.²⁷

The next step toward the spacetime metric that describes a uniform dust-filled universe comes from considering how a small distance is measured in two dimensions. For a flat plane, polar coordinates are an appropriate choice of variables, and the differential distance $d\ell$ between two nearby points P_1 and P_2 on the plane [see Fig. 29.18(a)] is given by

$$(d\ell)^2 = (dr)^2 + (r d\phi)^2.$$

Polar coordinates can also be used to measure the differential distance between two nearby points on the surface of a sphere. As an example, we will return to the surface of the

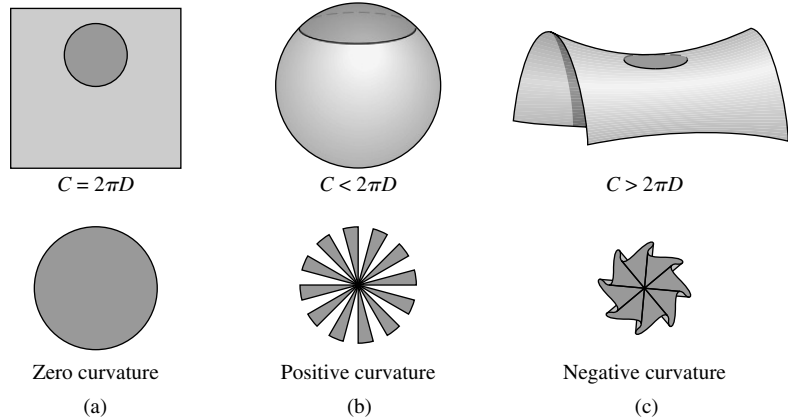


FIGURE 29.17 Calculating the curvature of a surface in three geometries: (a) a flat plane, (b) the surface of a sphere, and (c) the surface of a hyperboloid.

²⁷Unfortunately, it is not possible to visualize the negative curvature of the hyperboloid because its surface (unlike a sphere's) has infinite area. We cannot “step back” to view it as a whole.

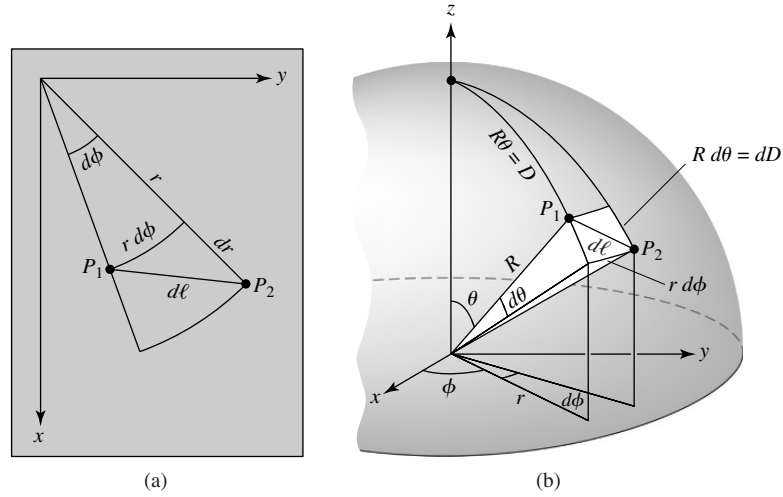


FIGURE 29.18 $d\ell$ as measured for (a) a flat plane and (b) the surface of a sphere.

sphere of radius R and curvature $K = 1/R^2$ considered previously. Then, as shown in Fig. 29.18(b), the distance $d\ell$ between two points P_1 and P_2 on the sphere is now given by

$$(d\ell)^2 = (dD)^2 + (r d\phi)^2 = (R d\theta)^2 + (r d\phi)^2.$$

But $r = R \sin \theta$, so $dr = R \cos \theta d\theta$ and

$$R d\theta = \frac{dr}{\cos \theta} = \frac{R dr}{\sqrt{R^2 - r^2}} = \frac{dr}{\sqrt{1 - r^2/R^2}}.$$

The differential distance on the sphere's surface may therefore be written as

$$(d\ell)^2 = \left(\frac{dr}{\sqrt{1 - r^2/R^2}} \right)^2 + (r d\phi)^2,$$

in terms of the plane polar coordinates r and ϕ . More generally, in terms of the curvature K of a two-dimensional surface,

$$(d\ell)^2 = \left(\frac{dr}{\sqrt{1 - Kr^2}} \right)^2 + (r d\phi)^2.$$

The extension to three dimensions is accomplished simply by making a transition from polar to spherical coordinates,

$$(d\ell)^2 = \left(\frac{dr}{\sqrt{1 - Kr^2}} \right)^2 + (r d\theta)^2 + (r \sin \theta d\phi)^2, \quad (29.104)$$

where r now measures the radial distance from the origin.

Equation (29.104) shows the effect of the curvature of our three-dimensional universe on spatial distances. Although the curvature of a two-dimensional spherical surface can be seen by stepping back from the sphere and viewing it immersed in its three-dimensional surroundings, there is unfortunately no fourth spatial dimension into which we can step back to view the curvature of our universe.

The final step toward the spacetime metric comes from recalling that by “distance,” we mean the *proper distance* between two spacetime events that occur simultaneously according to an observer. In an expanding universe, the positions of two points must be recorded at the same time if their separation is to have any meaning. In an isotropic, homogeneous universe, there is no reason why time should pass at different rates at different locations; consequently, the temporal term should simply be $c dt$.²⁸ If we take

$$(ds)^2 = (c dt)^2 - \left(\frac{dr}{\sqrt{1 - Kr^2}} \right)^2 - (r d\theta)^2 - (r \sin \theta d\phi)^2$$

as the metric for an isotropic, homogeneous universe, then Eq. (17.18) for the proper distance agrees with Eq. (29.104). That is, the differential proper distance is just $d\mathcal{L} = \sqrt{-(ds)^2}$ with $dt = 0$.

All that remains is to express this metric in terms of the dimensionless scale factor, $R(t)$, defined by $r(t) = R(t)\varpi$ (Eq. 29.3). Because the expansion of the universe affects all of its geometric properties, including its curvature, it is useful to define the time-dependent curvature in terms of a time-independent constant, k , as

$$K(t) \equiv \frac{k}{R^2(t)}. \quad (29.105)$$

These substitutions for r and K result in

$$(ds)^2 = (c dt)^2 - R^2(t) \left[\left(\frac{d\varpi}{\sqrt{1 - k\varpi^2}} \right)^2 + (\varpi d\theta)^2 + (\varpi \sin \theta d\phi)^2 \right], \quad (29.106)$$

which is known as the **Robertson–Walker metric**. The Robertson–Walker metric determines the spacetime interval between two events in an isotropic, homogeneous universe, just as the Schwarzschild metric, Eq. (17.22), is used to measure the interval between two events in the curved spacetime surrounding a massive object. The name honors Howard Percy Robertson (1903–1961) and Arthur Geoffrey Walker (1909–2001), who independently demonstrated, in the mid-1930s, that this is the most general metric possible for describing an isotropic and homogeneous universe.

In fact, we will use the same technique to define ϖ for a curved spacetime that we used in Section 17.2 to specify the radial coordinate r . From the Robertson–Walker metric, the area today ($t = t_0$ so $R(t_0) = 1$) of a spherical surface centered on the point $\varpi = 0$ is $4\pi \varpi^2$. By definition, this surface is located at the coordinate ϖ . It is important to remember that

²⁸You should compare this with the Schwarzschild metric, Eq. (17.22), where the presence of a central mass produces a more complicated time dependence.

the ϖ in Eq. (29.106) is a *comoving* coordinate that follows a given object as the universe expands. Furthermore, the time, t , is a *universal time* that essentially measures the time that has elapsed since the Big Bang. This is not an absolute time, but merely reflects a choice of how the clocks of distant observers are to be synchronized. For example, observers at various locations throughout the universe could, theoretically, synchronize their clocks using the cosmological principle by noting the precise temperature of the cosmic microwave background or the exact value of the Hubble constant.

The Friedmann Equation

Solving Einstein's field equations (Eq. 17.15) for an isotropic, homogeneous universe leads to a description of the dynamic evolution of the universe in the form of a differential equation for the scale factor, $R(t)$. This is known as the **Friedmann equation**,

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3} \pi G \rho \right] R^2 = -kc^2, \quad (29.107)$$

named for the Russian meteorologist and mathematician Aleksandr Friedmann (1888–1925).²⁹ In 1922 Friedmann solved Einstein's field equations for an isotropic, homogeneous universe to obtain this equation for a nonstatic universe. The same equation was derived independently in 1927 by the Belgian cleric Abbé Georges Lemaître (1894–1966).³⁰

The Cosmological Constant

Einstein realized that, as originally conceived, his field equations could not produce a static universe. Hubble's discovery of the expanding universe had not yet been made, so in 1917 Einstein modified his equations by adding an ad hoc term that contained the **cosmological constant**, Λ .³¹ With this addition, the general solution of Einstein's field equations is

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3} \pi G \rho - \frac{1}{3} \Lambda c^2 \right] R^2 = -kc^2. \quad (29.108)$$

Except for the cosmological constant, this is the same as the Friedmann equation. The additional term containing Λ would result from the Newtonian cosmology of Section 29.1 if a potential energy term,

$$U_\Lambda \equiv -\frac{1}{6} \Lambda m c^2 r^2,$$

²⁹According to general relativity, the density ρ includes the equivalent mass density of photons and neutrinos, as in Eq. (29.77).

³⁰Lemaître was the first person to propose that the present universe evolved from a highly dense beginning and so is sometimes called “the father of the Big Bang.”

³¹Some authors incorporate a factor of c^2 into the definition of Λ ; in this text, Λ has units of $(\text{length})^{-2}$.

were added to the left-hand side of Eq. (29.1). The conservation of mechanical energy applied to an expanding shell of mass m (described on page 1146) then becomes

$$\frac{1}{2}mv^2 - G\frac{M_r m}{r} - \frac{1}{6}\Lambda mc^2 r^2 = -\frac{1}{2}mkc^2 \varpi^2.$$

The force due to this new potential is

$$\mathbf{F}_\Lambda = -\frac{\partial U_\Lambda}{\partial r} \hat{\mathbf{r}} = \frac{1}{3}\Lambda mc^2 r \hat{\mathbf{r}} \quad (29.109)$$

(recall Eq. 2.15), which is radially outward for $\Lambda > 0$. In effect, a positive cosmological constant produces a repulsive force on the mass shell. This allowed Einstein to achieve his goal of balancing his static, closed universe against a gravitational collapse in an (unstable) equilibrium. Later, after the expansion of the universe had been discovered, Einstein expressed his regret at including the Λ -term in his field equations and has been quoted as referring to it as “the biggest blunder of my life.”

A nonzero cosmological constant implies that space would be curved even in an empty universe that is devoid of matter, an idea that Einstein disliked because it conflicted with his ideas concerning mass as the cause of spacetime curvature. Ironically, as mentioned in Section 27.2, Willem de Sitter used Einstein’s field equations with the Λ -term to describe an expanding, empty universe, with the expansion powered by the cosmological constant, and Hubble viewed the distance-dependent redshift of de Sitter’s universe as theoretical support for an expanding universe in his 1929 paper.

The Effects of Dark Energy

Nature has the final say, and in the late 1990s, observations forced astronomers to recognize a nonzero cosmological constant. Although we will defer a discussion of the physical origin of the cosmological constant to the next chapter, for the sake of discussion we will give the physical source of the cosmological constant a name: **dark energy**.³²

We begin by rewriting the Friedmann equation (including the Λ -term) in a form that makes it explicit that we are dealing with a three-component universe of mass (baryonic and dark), relativistic particles (photons and neutrinos), and dark energy,

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3}\pi G(\rho_m + \rho_{\text{rel}}) - \frac{1}{3}\Lambda c^2 \right] R^2 = -kc^2. \quad (29.110)$$

The **fluid equation** (Eq. 29.50) also emerges from solving Einstein’s field equations with the cosmological constant included, as

$$\frac{d(R^3 \rho)}{dt} = -\frac{P}{c^2} \frac{d(R^3)}{dt}, \quad (29.111)$$

where ρ and P are the density and pressure due to every component of the universe (including, as we shall see, dark energy). Note that although Λ was included in the field equations,

³²The term *dark energy* originated with cosmologist Michael Turner.

it does not appear in the fluid equation. As was done in Section 29.1, the Friedmann and fluid equations can be combined to produce the **acceleration equation**,

$$\frac{d^2 R}{dt^2} = \left\{ -\frac{4}{3}\pi G \left[\rho_m + \rho_{\text{rel}} + \frac{3(P_m + P_{\text{rel}})}{c^2} \right] + \frac{1}{3}\Lambda c^2 \right\} R. \quad (29.112)$$

(Although $P_m = 0$ for a pressureless dust universe, it is included in the acceleration equation for the sake of completeness.)

If we define the equivalent mass density of the dark energy to be

$$\rho_\Lambda \equiv \frac{\Lambda c^2}{8\pi G} = \text{constant} = \rho_{\Lambda,0}, \quad (29.113)$$

then the Friedmann equation becomes, in parallel with Eq. (29.65),

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3}\pi G(\rho_m + \rho_{\text{rel}} + \rho_\Lambda) \right] R^2 = -kc^2. \quad (29.114)$$

Because ρ_Λ remains constant as the universe expands, more and more dark energy must continually appear to fill the increasing volume.³³

The pressure due to dark energy, as calculated from Eq. (29.111), is

$$P_\Lambda = -\rho_\Lambda c^2. \quad (29.115)$$

Thus $w_\Lambda = -1$ in the general equation of state $P = w\rho c^2$ (Eq. 29.52). This equation of state is unlike any other we have encountered. A positive cosmological constant corresponds to a positive mass density and a negative pressure! Substituting the expressions for ρ_Λ and P_Λ into Eq. (29.112), the acceleration equation, yields

$$\frac{d^2 R}{dt^2} = \left\{ -\frac{4}{3}\pi G \left[\rho_m + \rho_{\text{rel}} + \rho_\Lambda + \frac{3(P_m + P_{\text{rel}} + P_\Lambda)}{c^2} \right] \right\} R. \quad (29.116)$$

With the inclusion of ρ_Λ and P_Λ , these equations have the same form as their Newtonian counterparts, Eqs. (29.10), (29.50), and (29.51). However, the interpretation of the constant k has changed. In Section 29.1, k was related to the mechanical energy of an expanding mass shell by Eq. (29.1). Here, it is seen to be the present value of the curvature of the universe [Eq. (29.105) with $R = 1$].³⁴

Using Eq. (29.8) and recalling from Eq. (29.12) that $3H^3/8\pi G = \rho_c$, the Friedmann equation can be written as

$$H^2 [1 - (\Omega_m + \Omega_{\text{rel}} + \Omega_\Lambda)] R^2 = -kc^2, \quad (29.117)$$

³³There are more general models in which the cosmological constant, Λ , is not really constant. In these models, dark energy is replaced by *quintessence* (“fifth element”), a time-dependent energy density. We will not pursue these models.

³⁴In some texts, the comoving coordinate, ϖ , is scaled so that k takes on the values of 0, +1, or −1 for a flat, closed, or open universe.

where

$$\Omega_{\Lambda} = \frac{\rho_{\Lambda}}{\rho_c} = \frac{\Lambda c^2}{3H^2}. \quad (29.118)$$

We define the **total density parameter** as³⁵

$$\Omega \equiv \Omega_m + \Omega_{\text{rel}} + \Omega_{\Lambda}. \quad (29.119)$$

The Friedmann equation then becomes

$$H^2(1 - \Omega)R^2 = -kc^2, \quad (29.120)$$

so for a flat universe ($k = 0$), we must have $\Omega(t) = 1$. It is useful to note that, as a special case at $t = t_0$,

$$H_0^2(1 - \Omega_0) = -kc^2. \quad (29.121)$$

Using Eqs. (29.120) and (29.121) along with Eqs. (29.79), (29.80), and (29.118), the Hubble parameter as a function of the redshift z is found to be

$$H = H_0(1+z) \left[\Omega_{m,0}(1+z) + \Omega_{\text{rel},0}(1+z)^2 + \frac{\Omega_{\Lambda,0}}{(1+z)^2} + 1 - \Omega_0 \right]^{1/2} \quad (29.122)$$

(cf. Eq. 29.27).

The WMAP values for $\Omega_{m,0}$, $\Omega_{\text{rel},0}$, and $\Omega_{\Lambda,0}$ are

$$\begin{aligned} [\Omega_{m,0}]_{\text{WMAP}} &= 0.27 \pm 0.04, \\ \Omega_{\text{rel},0} &= 8.24 \times 10^{-5}, \\ [\Omega_{\Lambda,0}]_{\text{WMAP}} &= 0.73 \pm 0.04, \end{aligned}$$

where the value of $\Omega_{\text{rel},0}$ was obtained from Eq. (29.80). Adding these results from the Wilkinson Microwave Anisotropy Probe reveals that

$$\Omega_0 = \Omega_{m,0} + \Omega_{\text{rel},0} + \Omega_{\Lambda,0} = 1;$$

that is, the universe is flat ($k = 0$), and dark energy now dominates the expansion of the universe. More precisely, the WMAP result is

$$[\Omega_0]_{\text{WMAP}} = 1.02 \pm 0.02,$$

which is *consistent* with $\Omega_0 = 1$. We can never prove the Ω_0 is *exactly* 1 because any measurement will necessarily be accompanied by a small (we hope) uncertainty.

³⁵The symbol “ Ω ” without a subscript “ m ,” “ rel ,” or “ Λ ” always denotes the total density parameter for all of the components of that particular model.

The deceleration parameter (Eq. 29.54) may be written as (Problem 29.42)

$$q(t) = \frac{1}{2} \sum_i (1 + 3w_i) \Omega_i(t), \quad (29.123)$$

where w is the coefficient from the equation of state $P_i = w_i \rho_i c^2$ and the “ i ” subscript identifies one of the components of the universe (i.e., pressureless dust, relativistic particles, or dark energy). Using $w_m = 0$, $w_{\text{rel}} = 1/3$, and $w_\Lambda = -1$, we obtain

$$q(t) = \frac{1}{2} \Omega_m(t) + \Omega_{\text{rel}}(t) - \Omega_\Lambda(t). \quad (29.124)$$

With WMAP values we find that the current value of the deceleration parameter is

$$q_0 = -0.60.$$

The minus sign indicates that expansion of the universe is now accelerating!³⁶

Assuming $\Lambda > 0$, the equivalent mass density of dark energy ρ_Λ adds to the effect of the other densities in the Friedmann equation (Eq. 29.114) on the curvature of the universe, k . However, in the acceleration equation, the negative pressure P_Λ opposes the gravitational effect of a positive ρ_Λ and acts to *increase* the acceleration of the universe, as seen by the positive term $\Lambda c^2/3$ added to the acceleration equation, Eq (29.112). The presence of a cosmological constant therefore decouples the geometry of the universe (open, closed, flat), which is described by k , from the dynamics of the universe, which are governed by the interplay of ρ_m , ρ_{rel} , and ρ_Λ . As we shall see, our universe may be flat ($k = 0$), yet it may have an accelerating expansion—a combination that is not possible within the more restricted model of a one-component universe of pressureless dust.

The Λ Era

The Λ -terms in Eqs. (29.114) and (29.116) became dominant as the scale factor R increased because ρ_Λ is constant while the mass density $\rho_m \propto R^{-3}$ and $\rho_{\text{rel}} \propto R^{-4}$. Just as the radiation era yielded to the mass era when the universe was about 55,000 years old (Eq. 29.85), the mass era has segued into the **Λ era**. Dark energy now governs the expansion of the universe. The transition from the matter era to the present Λ era occurred when the scale factor satisfied $\rho_m = \rho_\Lambda$. Since ρ_Λ is constant, Eq. (29.82) can be used to show that the transition value of R is

$$R_{m,\Lambda} = \left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} \right)^{1/3}. \quad (29.125)$$

Inserting WMAP values gives

$$R_{m,\Lambda} = 0.72, \quad (29.126)$$

³⁶By “accelerating,” we mean $d^2 R/dt^2 > 0$.

which corresponds to a redshift of

$$z_{m,\Lambda} = \frac{1}{R_{m,\Lambda}} - 1 = 0.39. \quad (29.127)$$

The acceleration equation with WMAP values shows (Problem 29.43) that the acceleration of the universe changed sign (from negative to positive) when the scale factor was

$$R_{\text{accel}} = 2^{-1/3} R_{m,\Lambda} = 0.57,$$

corresponding to

$$z_{\text{accel}} = 0.76,$$

meaning that the acceleration became positive *before* the Λ -term dominated the Friedmann equation. As $R \rightarrow 0$ we can deduce that the effects of the cosmological constant are negligible in the early universe because $\rho_m \propto R^{-3}$ and $\rho_{\text{rel}} \propto R^{-4}$ (Eqs. 29.81 and 29.82) while ρ_Λ remains constant. The happy outcome is that *all of the results for the early universe obtained in Sections 29.1 and 29.2 are valid for the present relativistic cosmology as well*; see Problem 29.40.

The behavior of the scale factor R for a flat universe can be found by setting $k = 0$ in the Friedmann equation (Eq. 29.114). A little algebra and Eqs. (29.79) and (29.118) lead to

$$t = \sqrt{\frac{3}{8\pi G}} \int_0^R \frac{R' dR'}{\sqrt{\rho_{m,0} R' + \rho_{\text{rel},0} + \rho_{\Lambda,0} R'^4}}. \quad (29.128)$$

Although this can be integrated numerically, it has no simple analytic solution. Figure 29.19 displays a numerical solution of Eq. (29.128) using WMAP values, showing the different behaviors of the scale factor in the radiation, matter, and Λ eras.

To make further progress, we will neglect the reign of relativistic particles during the first 55,000 years or so of the universe by setting $\rho_{\text{rel},0} = 0$. Integrating eventually yields, for $k = 0$,

$$t(R) = \frac{2}{3} \frac{1}{H_0 \sqrt{\Omega_{\Lambda,0}}} \ln \left[\sqrt{\left(\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}} \right) R^3 + 1} + \sqrt{\left(\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}} \right) R^3} \right]. \quad (29.129)$$

The present age of the universe may be obtained by substituting $R = 1$ into Eq. (29.129). Using the WMAP values for $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$, we obtain

$$t_0 = 4.32 \times 10^{17} \text{ s} = 1.37 \times 10^{10} \text{ yr.}$$

This is in good agreement with the best determination of the age of the universe currently available, the published WMAP value of³⁷

$[t_0]_{\text{WMAP}} = 13.7 \pm 0.2 \text{ Gyr.}$

³⁷When the first edition of this text was written, the authors could not have anticipated that the age of the universe, to this precision, would be included in the second edition!

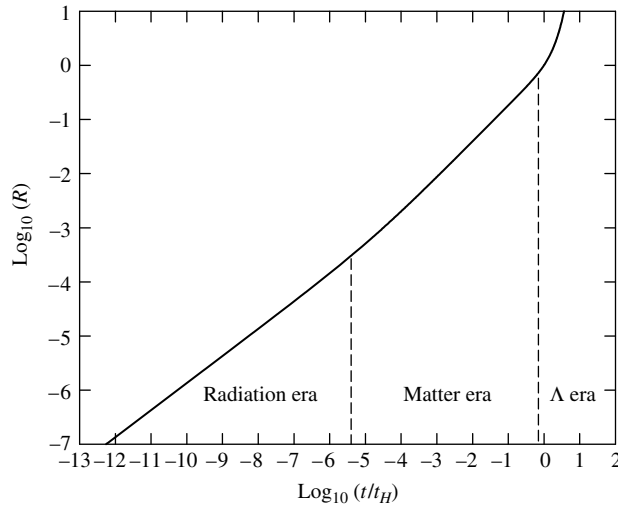


FIGURE 29.19 A logarithmic graph of the scale factor R as a function of time. During the radiation era, $R \propto t^{1/2}$; during the matter era, $R \propto t^{2/3}$; and during the Λ era, R grows exponentially.

The acceleration of the universe changed sign (from negative to positive) when $R = R_{\text{accel}} = 0.57$. According to Eq. (29.129), the expansion of the universe began to speed up when its age was

$$t_{\text{accel}} = 2.23 \times 10^{17} \text{ s} = 7.08 \text{ Gyr.}$$

Thus the expansion of the universe has been accelerating for approximately the second half of its existence. For this reason, t_0 is very nearly equal to the Hubble time:

$$t_0 = 0.993 t_H.$$

At the present epoch, the effects of deceleration during the radiation and matter eras and acceleration during the Λ era nearly cancel, so the age of the universe is what we would have calculated for a constant rate of expansion.

Equation (29.129) can be inverted to obtain, for $k = 0$,

$$R(t) = \left(\frac{\Omega_{m,0}}{4\Omega_{\Lambda,0}} \right)^{1/3} \left(e^{3H_0 t \sqrt{\Omega_{\Lambda,0}/2}} - e^{-3H_0 t \sqrt{\Omega_{\Lambda,0}/2}} \right)^{2/3} \quad (29.130)$$

$$= \left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} \right)^{1/3} \sinh^{2/3} \left(\frac{3}{2} H_0 t \sqrt{\Omega_{\Lambda,0}} \right) \quad (29.131)$$

Figure 29.20 shows the evolution of the scale factor as a function of time. In the limit of $3H_0 t \sqrt{\Omega_{\Lambda,0}/2} \ll 1$ (essentially when $t \ll t_H$), Eq. (29.131) reduces to Eq. (29.91),

$$R(t) \simeq \left(\frac{3}{2} H_0 t \sqrt{\Omega_{m,0}} \right)^{2/3} = \left(\frac{3\sqrt{\Omega_{m,0}}}{2} \right)^{2/3} \left(\frac{t}{t_H} \right)^{2/3}, \quad (29.132)$$

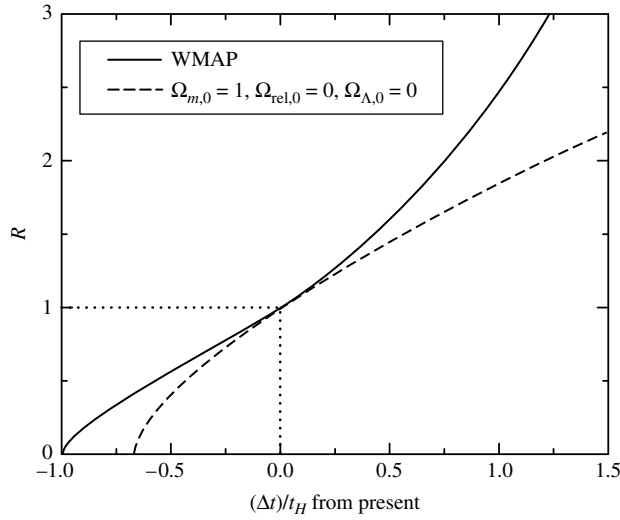


FIGURE 29.20 The scale factor R as a function of time, measured from the present, for a WMAP universe with $t_0 \simeq t_H$, and a flat, one-component universe of pressureless dust with $t_0 = 2t_H/3$ (Eq. 29.44). The dotted lines locate the position of today's universe on the two curves.

as appropriate for a matter-dominated universe. But when $t \gg t_H$, the second exponential in Eq. (29.131) can be neglected, leaving

$$R(t) \simeq \left(\frac{\Omega_{m,0}}{4\Omega_{\Lambda,0}} \right)^{1/3} e^{H_0 t \sqrt{\Omega_{\Lambda,0}}}. \quad (29.133)$$

When the cosmological constant dominates the Friedmann equation, the scale factor R grows exponentially with a characteristic time of $t_H/\sqrt{\Omega_{\Lambda,0}}$. As we will discover in the next section, an exponentially expanding universe has profound implications for the long-term future of observational astronomy.

Model Universes on the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ Plane

Every model of a three-component universe (matter, relativistic particles, and dark energy) is specified by the values of the three density parameters $\Omega_{m,0}$, $\Omega_{rel,0}$, and $\Omega_{\Lambda,0}$. At the present epoch $\Omega_{rel,0}$ is negligible, so we can consider a two-dimensional plot of $\Omega_{m,0}$ vs. $\Omega_{\Lambda,0}$, as shown in Fig. 29.21. The $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane is divided into several regions. The Friedmann equation (Eq. 29.117) shows that the line $\Omega_{m,0} + \Omega_{\Lambda,0} = 1$ determines the sign of k and so divides the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane into open and closed universes. The sign of the deceleration parameter (Eq. 29.123) is determined by the sign of the quantity $\Omega_{m,0} - 2\Omega_{\Lambda,0}$, so the line $\Omega_{m,0} - 2\Omega_{\Lambda,0} = 0$ divides the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane into accelerating and decelerating universes.

Although it seems that our universe will expand forever, driven by dark energy, it is easy to conceive of other model universes that will eventually recollapse. This includes any universe with $\Omega_{\Lambda,0} < 0$, as well as universes with $\Omega_{\Lambda,0} > 0$ but containing enough matter

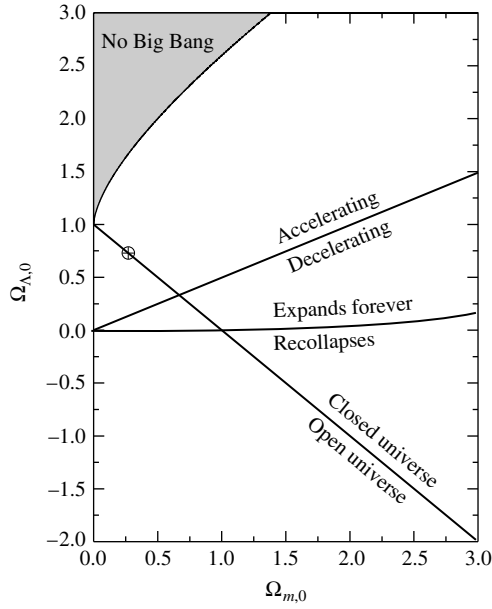


FIGURE 29.21 Model universes on the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane. Every point on this plane represents a possible universe. The point $(\Omega_{m,0} = 0.27, \Omega_{\Lambda,0} = 0.73)$ is indicated by the circle.

to bring the expansion to a halt before dark energy dominates. When the expansion stops, $dR/dt = 0$. Equation (29.114), together with Eqs. (29.81), (29.82), (29.113), and (29.15), can be used to express dR/dt as

$$\left(\frac{dR}{dt}\right)^2 = H_0^2 \left(\frac{\Omega_{m,0}}{R} + \frac{\Omega_{\text{rel},0}}{R^2} + \Omega_{\Lambda,0} R^2 + 1 - \Omega_{m,0} - \Omega_{\text{rel},0} - \Omega_{\Lambda,0} \right). \quad (29.134)$$

Setting the left-hand side equal to zero, canceling the H_0^2 , and neglecting the radiation era, we have a cubic equation for the scale factor R .

$$\frac{\Omega_{m,0}}{R} + \Omega_{\Lambda,0} R^2 + 1 - \Omega_{m,0} - \Omega_{\Lambda,0} = 0. \quad (29.135)$$

We want to know when this cubic equation for R has a positive, real root. It turns out that if $\Omega_{m,0} < 1$ and $\Omega_{\Lambda,0} > 0$, Eq. (29.135) has no positive, real root, and we conclude that these universes will expand forever.³⁸ But a universe with $\Omega_{m,0} > 1$ will expand forever only if

$$\Omega_{\Lambda,0} > 4\Omega_{m,0} \left\{ \cos \left[\frac{1}{3} \cos^{-1} \left(\frac{1}{\Omega_{m,0}} - 1 \right) + \frac{4\pi}{3} \right] \right\}^3.$$

³⁸You are referred to Felten and Isaacman (1986) for a complete analysis of this diagram.

On this diagram, lines of constant age t_0 are roughly diagonal from the lower left to the upper right, and t_0 increases from lower right to upper left. In fact, t_0 becomes *infinite* as we approach a line across the upper-left corner of the diagram where there is an (unstable) equilibrium between the inward pull of gravity and the outward push of dark energy. Models on this line are infinitely old, meaning that they never unfolded from a hot, dense Big Bang. This line is given by

$$\Omega_{\Lambda,0} = 4\Omega_{m,0} \left\{ \cos \left[\frac{1}{3} \cos^{-1} \left(\frac{1}{\Omega_{m,0}} - 1 \right) \right] \right\}^3,$$

if $\Omega_{m,0} > 0.5$; otherwise, “cos” should be replaced by “cosh.” Models beyond this line represent “bounce” universes that are now on the rebound from an earlier collapse. We merely state the result that these “bounce” models have a maximum redshift (at the bounce) that satisfies

$$z_{\text{bounce}} \leq 2 \cos \left[\frac{1}{3} \cos^{-1} \left(\frac{1 - \Omega_{m,0}}{\Omega_{m,0}} \right) \right] - 1 \quad (29.136)$$

if $\Omega_{m,0} > 0.5$; otherwise, “cos” should be replaced by “cosh.” Because objects are observed beyond this maximum redshift, these “bounce” models may be rejected.

Astronomers have the task of determining which point on Fig. 29.21 represents *our* universe. This task is possible because the dynamics of the expansion of the universe determines $q_0 = \Omega_{m,0}/2 - \Omega_{\Lambda,0}$, and the geometry of the universe determines $\Omega_0 = \Omega_{m,0} + \Omega_{\Lambda,0}$. In the next section we will learn how q_0 is measured, but the evaluation of Ω_0 will have to wait until Chapter 30.

29.4 ■ OBSERVATIONAL COSMOLOGY

Most of the key parameters of cosmology encountered in the previous section, such as H_0 , q_0 , and the various Ω_0 's, are not quantities that can be directly measured by astronomers. Observers are primarily limited to measuring the spectrum, redshift, radiant flux, and polarization of the starlight from a distant object. We now proceed to link these observables to the theoretical framework we have erected.

The Origin of the Cosmological Redshift

Let's begin by uncovering the origin of the cosmological redshift. We start with the Robertson–Walker metric, Eq. (29.106), with $ds = 0$ for a light ray (recall Section 17.2), and $d\theta = d\phi = 0$ for a radial path traveled from the point of the light's emission at comoving coordinate ϖ_e to its arrival at Earth at $\varpi = 0$. Taking the negative square root (so ϖ decreases with increasing time) gives

$$\frac{-c dt}{R(t)} = \frac{d\varpi}{\sqrt{1 - k\varpi^2}}.$$

Integrating this from a larger ϖ_{far} at an initial time t_i to a smaller ϖ_{near} at time t_f , we have

$$\int_{t_i}^{t_f} \frac{c dt}{R(t)} = - \int_{\varpi_{\text{far}}}^{\varpi_{\text{near}}} \frac{d\varpi}{\sqrt{1 - k\varpi^2}} = \int_{\varpi_{\text{near}}}^{\varpi_{\text{far}}} \frac{d\varpi}{\sqrt{1 - k\varpi^2}}. \quad (29.137)$$

A moment's thought reveals that the same result describes an outwardly moving light ray. Suppose that one crest of the light wave was emitted at time t_e and received at t_0 , and the next wave crest was emitted at $t_e + \Delta t_e$ and received at $t_0 + \Delta t_0$. These times, which describe how long it takes for the successive crests of the light wave to travel to Earth, satisfy the relations

$$\int_{t_e}^{t_0} \frac{c dt}{R(t)} = \int_0^{\varpi_e} \frac{d\varpi}{\sqrt{1 - k\varpi^2}} \quad (29.138)$$

for the first crest and

$$\int_{t_e + \Delta t_e}^{t_0 + \Delta t_0} \frac{c dt}{R(t)} = \int_0^{\varpi_e} \frac{d\varpi}{\sqrt{1 - k\varpi^2}} \quad (29.139)$$

for the next. The right-hand sides are the same, since the comoving coordinate of an object does not change as the universe expands (assuming its peculiar velocity is negligible). Subtracting Eq. (29.138) from Eq. (29.139) produces

$$\int_{t_e + \Delta t_e}^{t_0 + \Delta t_0} \frac{dt}{R(t)} - \int_{t_e}^{t_0} \frac{dt}{R(t)} = 0. \quad (29.140)$$

But

$$\int_{t_e + \Delta t_e}^{t_0 + \Delta t_0} \frac{dt}{R(t)} = \int_{t_e + \Delta t_e}^{t_e} \frac{dt}{R(t)} + \int_{t_e}^{t_0} \frac{dt}{R(t)} + \int_{t_0}^{t_0 + \Delta t_0} \frac{dt}{R(t)},$$

so

$$\int_{t_0}^{t_0 + \Delta t_0} \frac{dt}{R(t)} - \int_{t_e}^{t_e + \Delta t_e} \frac{dt}{R(t)} = 0.$$

Any change in $R(t)$ during the time intervals Δt_e and Δt_0 between the emission of the two successive wave crests can safely be neglected. This allows us to treat $R(t)$ as a constant with respect to the time integration, so that, using $R(t_0) = 1$,

$$\Delta t_0 = \frac{\Delta t_e}{R(t_e)}. \quad (29.141)$$

The times Δt_e and Δt_0 are just the periods of the emitted and received light waves and are related to their wavelengths by $\lambda = c\Delta t$. Making this substitution into Eq. (29.141) and using the definition of the redshift z (Eq. 4.34) results in the expression for the cosmological redshift,

$$\frac{1}{R(t_e)} = \frac{\lambda_0}{\lambda_e} = 1 + z. \quad (29.142)$$

This derivation shows that the cosmological redshift is due to the wavelength of a photon expanding along with the space through which the photon moves during its journey to Earth. Equation (29.142) is just Eq. (29.4) discussed previously (this result was also discussed in the context of the distances to QSOs on page 1100). Combining Eqs. (29.141) and (29.142) results in the formula for **cosmological time dilation**,

$$\frac{\Delta t_0}{\Delta t_e} = 1 + z. \quad (29.143)$$

Note that these relations for the cosmological redshift and time dilation hold regardless of the functional form of the scale factor, $R(t)$. Experimental confirmation of cosmological time dilation has been frustrated by the lack of a reliable natural clock located at cosmological distances. However, a measurement of cosmological time dilation has been made using the temporal changes in the spectrum of a moderate-redshift ($z = 0.361$) Type Ia supernova. The results were consistent with Eq. (29.143). [See Foley et al. (2005) for details of this measurement of cosmological time dilation.]

Distances to the Most Remote Objects in the Universe

To make further progress in our quest to survey the universe and determine its geometry and dynamics, we must next learn how to measure the distance to objects in the most remote regions of the universe. The proper distance of an object from Earth can be found from the Robertson–Walker metric. Recall that the differential proper distance (Eq. 17.18) is just $\sqrt{-(ds)^2}$ with $dt = 0$. Furthermore, if the comoving coordinate of the object is ϖ (with Earth at $\varpi = 0$), then $d\theta = d\phi = 0$ along a radial line from Earth to the object. Inserting these into the Robertson–Walker metric (Eq. 29.106), we can find the proper distance, $d_p(t)$, to the object at time t by integrating

$$d_p(t) = R(t) \int_0^{\varpi} \frac{d\varpi'}{\sqrt{1 - k\varpi'^2}}. \quad (29.144)$$

Using Eq. (29.138), this becomes

$$d_p(t) = R(t) \int_{t_e}^{t_0} \frac{c dt'}{R(t')}. \quad (29.145)$$

The physical meaning of this is readily apparent. As the photon moves in from ϖ_e , in each interval of time dt it travels a small distance of $c dt$. These intervals cannot simply be added up, because the universe expands as the photon travels. Dividing $c dt$ by the scale factor at the time, $R(t)$, converts this small distance to what it would be at the present time, t_0 . Integrating then yields the proper distance from ϖ_e to $\varpi = 0$ today, at time t_0 . Multiplying by the scale factor $R(t)$ then converts this to the distance at some other time t . It is worth emphasizing that the current value of the proper distance, $d_{p,0} \equiv d_p(t_0)$, to an object is how far away it is *today*, and not its distance when its light was emitted. As long as the object has zero peculiar velocity (constant comoving coordinate ϖ), it suffices to find $d_{p,0}$ because

the proper distance at any other time can be obtained from

$$d_p(t) = R(t)d_{p,0}. \quad (29.146)$$

In particular, if the object's redshift is z , then its distance at time t_e , when its light was emitted, is

$$d_p(t_e) = d_{p,0}R(t_e) = \frac{d_{p,0}}{1+z}. \quad (29.147)$$

Integrating Eq. (29.144) and using $R(t_0) = 1$ shows that the expression for the present proper distance in a flat universe is

$$d_{p,0} = \varpi \quad (\text{for } k = 0), \quad (29.148)$$

the expression for a closed universe is

$$d_{p,0} = \frac{1}{\sqrt{k}} \sin^{-1}(\varpi\sqrt{k}) \quad (\text{for } k > 0), \quad (29.149)$$

and the expression for an open universe is

$$d_{p,0} = \frac{1}{\sqrt{|k|}} \sinh^{-1}(\varpi\sqrt{|k|}) \quad (\text{for } k < 0). \quad (29.150)$$

In a flat universe, the present proper distance to an object is just its coordinate distance $d_{c,0} = \varpi$ (cf. Eq. 29.3). However, the coordinate distance will not agree with the proper distance if $k \neq 0$. Because $\sin^{-1}(x) \geq x$, in a closed universe ($k > 0$) the proper distance to an object is *greater* than its coordinate distance. Similarly, $\sinh^{-1}(x) \leq x$, so in an open universe ($k < 0$) the proper distance to an object is *less* than its coordinate distance. Later we will find expressions for the proper distance $d_{p,0}$ and the comoving coordinate ϖ as functions of the redshift z . First, however, we pause to examine the above equations for $d_{p,0}$ because measuring distances in an expanding universe has some interesting aspects.

Distances in a closed universe ($k > 0$) are especially interesting. Solving Eq. (29.149) for ϖ gives

$$\varpi = \frac{1}{\sqrt{k}} \sin(d_{p,0}\sqrt{k}). \quad (29.151)$$

Note that in a closed universe there is a maximum value of the comoving coordinate of $\varpi_{\max} = 1/\sqrt{k}$. Also, there are an infinite number of distances along a radial line to the same point X in space, located, say, at ϖ_X . If $d_{p,0}$ is one value of the present proper distance to X at time t , then for any integer n , $d_{p,0} + 2\pi n/\sqrt{k}$ will also bring us back to X , with the same value of ϖ_X . The extra multiples of $2\pi/\sqrt{k}$ correspond to traversing the circumference of the universe n times before stopping at X . However, such a journey, running circles around the universe, would not be physically possible. As it happens, a photon emitted at $t = 0$ in a closed, matter-dominated universe with $\Lambda = 0$ would return to its starting point just as the universe ended in a Big Crunch; see Problem 29.47.

In a similar manner, our tiny ant that lives on the surface of a sphere could walk from pole to pole to pole, encircling the sphere n times before proceeding on to its final destination. This illustrates that although there is no boundary to a closed universe, it contains only a finite amount of space, like the unbounded surface of a sphere. Furthermore, a closed universe curves back on itself; moving outward from Earth (or from any other choice of origin), the farthest you can get from home is a point where $\varpi = 1/\sqrt{k}$. From that point, a step in *any* direction brings you closer to where you started.

You should be warned that suggestive phrases like “the circumference of the universe” and “curves back on itself” do not imply a curved path through three-dimensional space, since there is no deviation from a radial line as ϖ increases. Despite these caveats, we can define the circumference of a closed universe (including the time-dependence) as

$$C_{\text{univ}}(t) = \frac{2\pi R(t)}{\sqrt{k}} \quad (29.152)$$

which is the proper distance along a radial line that brings you back to your starting point. This expression for the circumference is consistent with our definition of curvature since, from Eq. (29.105), the radius of curvature of a closed universe at time t is $1/\sqrt{K(t)} = R(t)/\sqrt{k}$. Nevertheless, the radius of curvature must not be thought of as the radius of an actual circular path.

The Particle Horizon and the Horizon Distance

As the universe expands and ages, photons from increasingly distant objects have more time to complete their journey to Earth. This means that as time increases, we might expect that more of the universe will come into causal contact with the observer. The proper distance to the farthest observable point (called the **particle horizon**) at time t is the **horizon distance**, $d_h(t)$. Note that two points separated by a distance greater than d_h are not in causal contact. Thus d_h may be thought of as the *diameter* of the largest causally connected region.

We will now derive an expression for $d_h(t)$, the size of the observable universe as a function of time. (It is important to note that because the farthest observable point moves outward through increasingly larger values of ϖ , $d_h(t)$ is *not* proportional to $R(t)$.) Consider an observer at the origin ($\varpi = 0$), and let the particle horizon for this observer be located at ϖ_e at time t . This means that a photon emitted at ϖ_e at $t = 0$ would reach the origin at time t . With an appropriate change of limits in Eq. (29.145), the horizon distance at time t is found to be

$$d_h(t) = R(t) \int_0^t \frac{c dt'}{R(t')}. \quad (29.153)$$

First we consider distances in the early universe, when the effect of dark energy was negligible (so Λ may be set equal to zero). During the radiation era, the universe was essentially flat and the scale factor was of the form $R(t) = Ct^{1/2}$, where C is a constant (see Eq. 29.87). Inserting this into Eq. (29.153) gives

$$d_h(t) = 2ct \quad (\text{radiation era}). \quad (29.154)$$

After the radiation era, the expansion of the universe was governed by the effects of matter and, later, dark energy.

For the matter era, assuming a flat universe, the scale factor is given by Eq. (29.91), which is of the form $R(t) = Ct^{2/3}$, where again C is a constant. (Since the radiation era lasted only 55,000 years following the Big Bang, for the purposes of the following calculations we will ignore radiation and set the lower limit to $t = 0$.) Substituting this into Eq. (29.153) results in

$$d_h(t) = 3ct \quad (\text{for } k = 0). \quad (29.155)$$

Using Eqs. (29.4) and (29.90), this expression can be rewritten in terms of the redshift as

$$d_h(z) = \frac{2c}{H_0\sqrt{\Omega_{m,0}}} \frac{1}{(1+z)^{3/2}} \quad (\text{for } k = 0). \quad (29.156)$$

We can obtain a rough estimate for the present horizon distance by setting $z = 0$ to obtain

$$d_{h,0} \approx \frac{2c}{H_0\sqrt{\Omega_{m,0}}} = 5.02 \times 10^{26} \text{ m} = 16,300 \text{ Mpc} = 16.3 \text{ Gpc} \quad (29.157)$$

using WMAP values.

Finally, in the Λ era, we substitute Eq. (29.131) for the scale factor into Eq. (29.153) for the horizon distance to get, for $k = 0$,

$$d_h(t) = \left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} \right)^{1/3} \sinh^{2/3} \left(\frac{3}{2} H_0 t \sqrt{\Omega_{\Lambda,0}} \right) \int_0^t \frac{c dt'}{\left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} \right)^{1/3} \sinh^{2/3} \left(\frac{3}{2} H_0 t' \sqrt{\Omega_{\Lambda,0}} \right)}. \quad (29.158)$$

This has no simple analytic solution and must be integrated numerically. Using $t_0 = 13.7$ billion years, we calculate that at the present time, the distance to the particle horizon in a flat universe is

$$d_{h,0} = 4.50 \times 10^{26} \text{ m} = 14,600 \text{ Mpc} = 14.6 \text{ Gpc} \quad (29.159)$$

Figure 29.22 uses WMAP values and shows d_h , the size of the observable universe, as a function of time. Of course, when viewing an object near the particle horizon, astronomers see it as it was when the light was emitted, not as it would appear in today's universe.

Note that the distance to the particle horizon in Eqs. (29.154) and (29.155) is proportional to t , while the scale factor in the radiation and matter eras is proportional to $t^{1/2}$ and $t^{2/3}$, respectively. This means that during those eras the size of the observable universe increased more rapidly than the universe expanded, so the universe became increasingly causally connected as it aged. However, the integral in Eq. (29.158)—without the term in front—is just the present distance to the point that will be at the particle horizon at time t , as we can see by comparing Eq. (29.153) with Eq. (29.145), evaluated at t_0 with $t_e = 0$. As $t \rightarrow \infty$, this integral converges to 19.3 Gpc. This means that the proper distance *today* to the farthest

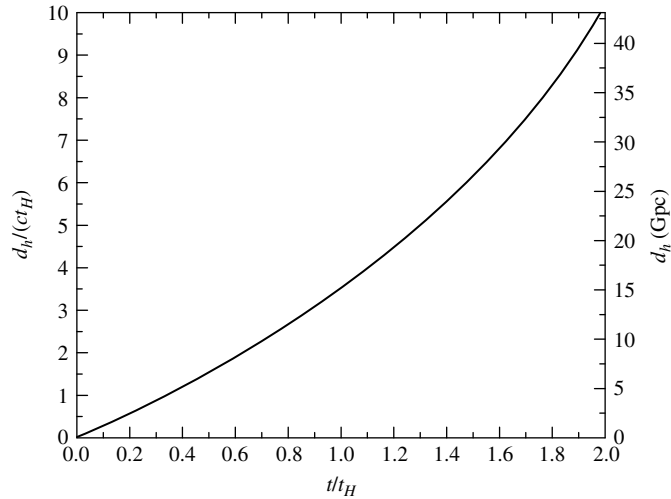


FIGURE 29.22 The proper distance from Earth to the particle horizon as a function of time, using WMAP values. The horizon distance is expressed as a fraction of ct_H . The right axis shows d_h in billions of parsecs.

object that will *ever* be observable in the future is 19.3 Gpc. Everything within a sphere, centered on Earth, of radius 19.3 Gpc will eventually become visible, while everything beyond will be forever hidden. In the future, the particle horizon and the scale factor will both grow exponentially as $e^{H_0 t \sqrt{\Omega_{\Lambda,0}}}$ (Eq. 29.133).

Ultimately, an object located at the particle horizon will remain at the particle horizon as the universe expands. The particle horizon will never catch up to any object that is presently more than 19.3 Gpc away, and so its light will *never* reach us. What will we observe when looking at an object at the ultimate ($t \rightarrow \infty$) particle horizon? Although photons from the object will continue to arrive, they will be increasingly deeply redshifted, and their arrival rate will decline toward zero due to cosmological time dilation (Eq. 29.143). Thus the object will fade from view, apparently frozen in time, as its redshift diverges to infinity. This bears a striking similarity to the description in Section 17.3 of how we would view an astronaut falling into a black hole, although the physics of the two situations is completely different.

Example 29.4.1. Helium nuclei were being formed when the temperature was roughly 10^9 K and $t = 178$ s. From Eq. (29.58), the scale factor at that time was $R = 2.73 \times 10^{-9}$. From Eq. (29.154), the horizon distance was then about

$$d_h(t) = 2ct = 1.07 \times 10^{11} \text{ m} = 0.7 \text{ AU}.$$

This is the diameter of a causally connected region, which we call $\mathcal{C}$, when the universe was 178 s old.

The region $\mathcal{C}$ (which has a comoving boundary, so it always contains the same mass) has been expanding along with the rest of the universe since $t = 178$ s. How large is $\mathcal{C}$

continued

today? Assuming a flat universe, Eq. (29.146) shows that $\mathcal{C}$ has expanded by a factor of $1/R = 3.66 \times 10^8$, with a present diameter of

$$\frac{d_h(t)}{R(t)} = 3.92 \times 10^{19} \text{ m},$$

about 1.3 kpc. In other words, the largest region that was causally connected at $t = 178$ s is now just over a kiloparsec across, roughly 8.7×10^{-8} of the present horizon distance, $d_{h,0}$. This illustrates that as the early universe aged, the amount of material that is in causal contact increased dramatically. Today's causally connected universe extends far beyond $\mathcal{C}$ because, since $t = 178$ s, light from more distant regions has had time to arrive and causally connect those regions with $\mathcal{C}$. The comoving boundary of $\mathcal{C}$ cannot keep up with the more rapid recession of the particle horizon.

The Arrival of Photons

You may be wondering, if the scale factor R was zero at the Big Bang and everything was right next to everything else, then why has it taken the age of the universe for a Big Bang photon to reach Earth? What is the path followed by the photon?

In the following discussion, the actual complications of the Big Bang will be neglected. Instead, we will consider a perfectly transparent, expanding, flat universe where a single photon is emitted at comoving coordinate ϖ_e at time $t = 0$. What, then, is the proper distance of that photon from our position ($\varpi = 0$) at a later time t ? The coordinate, ϖ , of the photon at time t may be found from Eq. (29.137) with $k = 0$,

$$\int_0^t \frac{c dt'}{R(t')} = \int_{\varpi}^{\varpi_e} d\varpi'. \quad (29.160)$$

To simplify the calculation, we will ignore the Λ era and adopt a flat, matter-dominated universe with a scale factor given by Eq. (29.132),

$$R(t) = \left(\frac{3}{2} H_0 t \sqrt{\Omega_{m,0}} \right)^{2/3}. \quad (29.161)$$

Setting $R = 1$, the age of this model universe is found to be

$$t_0 = \frac{2}{3 H_0 \sqrt{\Omega_{m,0}}}, \quad (29.162)$$

so the scale factor is simply

$$R(t) = \left(\frac{t}{t_0} \right)^{2/3}.$$

Integrating Eq. (29.160) yields

$$\varpi = \varpi_e - 3ct_0 \left(\frac{t}{t_0} \right)^{1/3}. \quad (29.163)$$

We can evaluate ϖ_e by noting that at $t = t_0$, $\varpi = 0$; thus

$$\varpi_e = 3ct_0, \quad (29.164)$$

the present horizon distance for this model (Eq. 29.155). Inserting this value for the photon's starting point into Eq. (29.163) and multiplying both sides by the scale factor $R(t)$ shows that the proper distance of the photon from Earth is, as a function of time,

$$d_p(t) = 3ct_0 \left[\left(\frac{t}{t_0} \right)^{2/3} - \left(\frac{t}{t_0} \right) \right] \quad (29.165)$$

for our model flat universe.

Since the Big Bang, the entire system of comoving coordinates has been stretching out from its compact beginning. Indeed, the “Big Stretch” would be a more appropriate term than the “Big Bang” for this event! As shown in Fig. 29.23, the initial expansion of the universe actually carried the photon *away* from Earth. Although the photon's comoving coordinate was always decreasing from an initial value of ϖ_e toward Earth's position at $\varpi = 0$, the scale factor $R(t)$ increased so rapidly that at first the proper distance between the photon and Earth increased with time. This means that a photon emitted from the present particle horizon at $t = 0$ is only now reaching Earth. Photons emitted from a greater ϖ , beyond the present particle horizon, have yet to arrive—and in fact may *never* arrive if ϖ is sufficiently large that the exponential expansion of the universe ultimately carries the photon away from Earth.

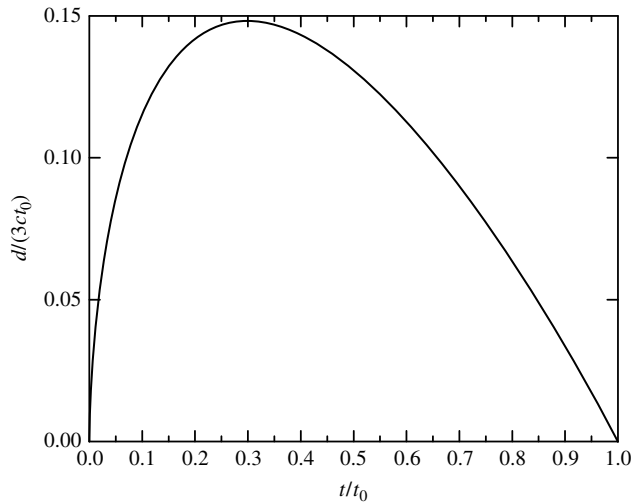


FIGURE 29.23 The proper distance from Earth of a photon emitted from the present particle horizon at the time of the Big Bang. The photon's proper distance is expressed as a fraction of $3ct_0$.

The Maximum Visible Age of a Source

The previous calculation assumes that the photon was emitted at $t = 0$. Is it possible that the exponential expansion of space could carry a presently visible object away from Earth so fast that the object would never again be seen in the sky from some future time forward? To answer this, consider an object (say, a galaxy) that is now visible, meaning that its light was emitted at an earlier time t_e and it arrives today at t_0 . Assuming that the galaxy is still visible at some time in the future, we will let the time of the emission of its future photons be t_i and their arrival time here be t_f , where $t_e < t_i$ and $t_0 < t_f$. Applying these conditions to Eq. (29.140) gives

$$\int_{t_e}^{t_0} \frac{dt}{R(t)} = \int_{t_i}^{t_f} \frac{dt}{R(t)}, \quad (29.166)$$

where the scale factor R is given by Eq. (29.131) for a flat universe ($k = 0$). Because the scale factor increases monotonically, it may be, for sufficiently large t_i , that *no* value of t_f can satisfy this equality. In that case, a photon emitted at time t_i will never reach Earth. The latest time of emission, t_{mva} , for photons to eventually reach us (the **maximum visible age** of the source) may be found by setting $t_f = \infty$.³⁹

Just like an object located at the ultimate particle horizon, as photons from this galaxy continue to arrive, they will be increasingly deeply redshifted and their flux will drop toward zero. The galaxy will fade from view, apparently frozen in time, as its redshift diverges to infinity. The farther a source is, the sooner it will fade away. This places a fundamental limit on extragalactic astronomy. We will never be able to watch galaxies continue to age and evolve through the millenia. Figure 29.24 shows that if the redshift of an object is roughly larger than 1.8, then $t_{\text{mva}} < t_H$, and we will never see it even as it appears *today*. That is, the light emitted by the object today will never reach Earth because those photons will eventually be carried away from us by the accelerating Hubble flow. Objects with a redshift in the range 5–10 can be observed only as they appeared when the universe

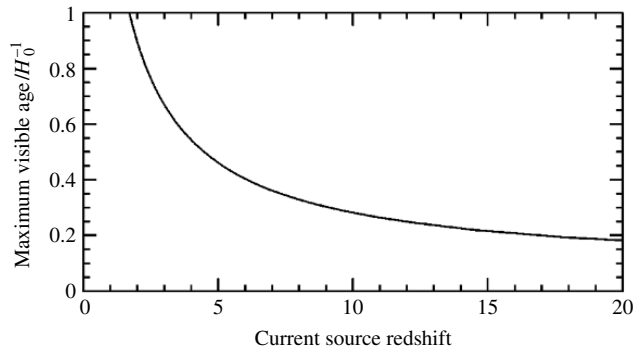


FIGURE 29.24 The maximum visible age of a source, in units of $t_H = 1/H_0$, as a function of its current redshift. (Figure adapted from Loeb, *Phys. Rev. D*, 65, 047301, 2002.)

³⁹Actually, t_{mva} is the age of the universe when the source emitted the last photons to reach Earth.

was approximately 4 to 6 billion years old. As more light is carried away from Earth by the expansion of the universe, the observable sky will become increasingly empty. Similarly, any signal we send toward a galaxy with $z \approx 1.8$ or more will never arrive. Because there can be no contact between that galaxy and Earth anytime in the future, we are no longer in causal contact with it. As the universe ages, it is becoming causally fragmented, with one region no longer capable of influencing another.⁴⁰

The Comoving Coordinate $\varpi(z)$

Returning to Eqs. (29.148–29.150), we want to express the comoving coordinate ϖ as a function of the redshift z . We begin by using Eq. (29.145) to find another expression for the present proper distance $d_{p,0}$. Writing $dt = dR/(dR/dt)$, Eq. (29.145) may be written as

$$d_{p,0} = \int_{R(t_e)}^{R(t_0)} \frac{c dR}{R(dR/dt)}.$$

It is useful to use $R(t_0) = 1$, $R(t_e) = 1/(1+z)$, $dR = -R^2 dz$ [from differentiating Eq. (29.4), $R = 1/(1+z)$] and Eq. (29.8) to define the dimensionless integral

$$I(z) = H_0 \int_{\frac{1}{1+z}}^1 \frac{dR}{R(dR/dt)} = H_0 \int_0^z \frac{dz'}{H(z')}. \quad (29.167)$$

Using Eq. (29.122), we obtain

$$I(z) \equiv \int_0^z \frac{dz'}{\sqrt{\Omega_{m,0}(1+z')^3 + \Omega_{rel,0}(1+z')^4 + \Omega_{\Lambda,0} + (1-\Omega_0)(1+z')^2}}. \quad (29.168)$$

With this definition of the integral $I(z)$, the present proper distance is

$$d_{p,0}(z) = \frac{c}{H_0} I(z). \quad (29.169)$$

Comparing this with Eqs. (29.148–29.150) and using Eq. (29.121) for k , we find our expressions for the comoving coordinate $\varpi(z)$:

$$\varpi(z) = \frac{c}{H_0} I(z) \quad (\Omega_0 = 1) \quad (29.170)$$

$$= \frac{c}{H_0 \sqrt{\Omega_0 - 1}} \sin \left[I(z) \sqrt{\Omega_0 - 1} \right] \quad (\Omega_0 > 1) \quad (29.171)$$

$$= \frac{c}{H_0 \sqrt{1 - \Omega_0}} \sinh \left[I(z) \sqrt{1 - \Omega_0} \right] \quad (\Omega_0 < 1). \quad (29.172)$$

⁴⁰Recall that gravitationally bound systems do not participate in the expansion of the universe, so our Solar System and Galaxy will not become causally fragmented.

These exact expressions must be evaluated numerically. For later reference, we define

$$S(z) \equiv I(z) \quad (\Omega_0 = 1) \quad (29.173)$$

$$\equiv \frac{1}{\sqrt{\Omega_0 - 1}} \sin \left[I(z) \sqrt{\Omega_0 - 1} \right] \quad (\Omega_0 > 1) \quad (29.174)$$

$$\equiv \frac{1}{\sqrt{1 - \Omega_0}} \sinh \left[I(z) \sqrt{1 - \Omega_0} \right] \quad (\Omega_0 < 1), \quad (29.175)$$

so we may simply write

$$\varpi(z) = \frac{c}{H_0} S(z). \quad (29.176)$$

Note that because $\sin(x) = x - x^3/3! + x^5/5! + \dots$ and $\sinh(x) = x + x^3/3! + x^5/5! + \dots$ we have the useful approximation (to second order in z) that $S(z) \simeq I(z)$. Thus

$$\varpi(z) \simeq \frac{c}{H_0} I(z). \quad (\text{for } z \ll 1) \quad (29.177)$$

Because the comoving coordinate ϖ is so important in observational cosmology, it will be helpful to find an approximate expression for the integral $I(z)$. (Again we will ignore the brief radiation era, so $\Omega_{\text{rel},0} = 0$ and $\Omega_0 = \Omega_{m,0} + \Omega_{\Lambda,0}$.) The integrand may be expressed as a Taylor series about $z = 0$ as

$$I(z) = \int_0^z \left\{ 1 - (1 + q_0)z' + \left[\frac{1}{2} + 2q_0 + \frac{3}{2}q_0^2 + \frac{1}{2}(1 - \Omega_0) \right] z'^2 + \dots \right\} dz' \quad (29.178)$$

where we have used (Eq. 29.124) for the deceleration parameter, $q_0 = \frac{1}{2}\Omega_{m,0} - \Omega_{\Lambda,0}$. Integrating gives our result,

$$I(z) = z - \frac{1}{2}(1 + q_0)z^2 + \left[\frac{1}{6} + \frac{2}{3}q_0 + \frac{1}{2}q_0^2 + \frac{1}{6}(1 - \Omega_0) \right] z^3 + \dots \quad (29.179)$$

Eqs. (29.170–29.172) then provide series expressions for the comoving coordinate ϖ of an object observed at a redshift z .

Note that the squared term in Eq. (29.179) involves only q_0 and thus depends only on the dynamics of the expanding universe, while the cubed term involves both q_0 and k (through Eq. 29.121) and so depends on both the dynamics and the geometry of the cosmos. A further simplification comes from using only the first two terms of the series expression for $I(z)$ along with Eq (29.177) to obtain, to second order in z ,

$$\varpi \simeq \frac{cz}{H_0} \left[1 - \frac{1}{2}(1 + q_0)z \right] \quad (\text{for } z \ll 1). \quad (29.180)$$

Equation (29.180) is valid regardless of whether or not the universe is flat and whether or not the cosmological constant, Λ , is nonzero. In fact, Eq. (29.180) can be derived very generally, without reference to the Friedmann equation or any specific model of the universe. The procedure, which is outlined in Problem 29.52, uses the fact that the deceleration parameter is defined as a second time derivative of the scale factor; recall Eq. (29.54).

The Proper Distance

We have arrived at our goal, an approximate expression for the proper distance of an object *at the present time*. According to Eq. (29.169), this is

$$d_{p,0} \simeq \frac{cz}{H_0} \left[1 - \frac{1}{2} (1 + q_0)z \right] \quad (\text{for } z \ll 1). \quad (29.181)$$

The first term is just the Hubble law, after applying Eq. (4.39) for the redshift. Since $q_0 = \Omega_{m,0}/2 - \Omega_{\Lambda,0}$, we see from the second term that larger values of $\Omega_{m,0}$ imply smaller distances (more mass to slow down the expansion of the universe), as do smaller values of $\Omega_{\Lambda,0}$ (less dark energy pressure to speed up the expansion). As we shall see, the second term involves a departure from the linearity of the Hubble law that can be used to determine the deceleration parameter, q_0 . For $q_0 = -0.6$, the second term is 10% of the first when $z = 0.13$. This is the reason for the statement on page 1056 that Eq. (27.8) should be abandoned for values of z greater than 0.13.

The Luminosity Distance

Now we are ready for the concept of the luminosity distance (as measured by the inverse square law, Eq. 3.2). Then we will be prepared to describe some of the classic observational tests of cosmological models. First, we associate the source of the emitted photons with the rate at which energy arrives at a telescope's detectors. Suppose that a radiant flux F is measured for a source of light with a known luminosity L . (For now, we will assume that F is a bolometric flux, measured over all wavelengths.) Then the inverse square law, Eq. (3.2), can be used to define the **luminosity distance**, d_L , of the star by

$$d_L^2 \equiv \frac{L}{4\pi F}. \quad (29.182)$$

Let a source of light be located at the origin ($\varpi = 0$) of a comoving coordinate system. The source emits photons that arrive at a spherical surface around the origin for which $\varpi = \text{constant} > 0$. From the Robertson–Walker metric, Eq. (29.106), the surface area of the sphere at the present time ($R = 1$) is $4\pi\varpi^2$; see Fig. 29.25.⁴¹ After traveling out to ϖ from the source, the photons will be spread over this surface area, and so the radiant flux will diminish as $1/\varpi^2$. Two effects, in addition to the inverse square law, act to further reduce the value of the radiant flux measured at this sphere. The cosmological redshift, Eq. (29.142), shows that the energy of each photon, $E_{\text{photon}} = hc/\lambda$, is reduced by a factor of $1 + z$. Also, cosmological time dilation, Eq. (29.143), affects the average time interval *between* photons emitted by the source. This means that the rate at which the photons arrive at the sphere is less than the rate at which they leave the source by another factor of $1 + z$. Combining these effects, the radiant flux at the sphere's surface is

$$F = \frac{L}{4\pi\varpi^2(1+z)^2}.$$

⁴¹ We emphasize that although the surface area of a sphere centered at the origin is $4\pi[R(t)\varpi]^2$, the proper distance of the surface from the origin is *not* $R(t)\varpi$ when $k \neq 0$; see Eqs. (29.149) and (29.150).

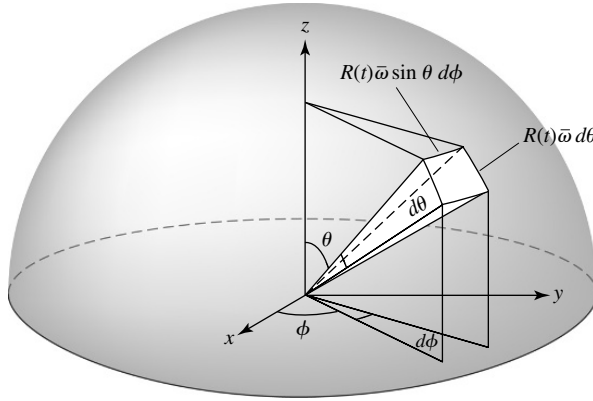


FIGURE 29.25 An element of area on the surface of a sphere centered at $\varpi = 0$. Integrating over the angles θ and ϕ shows that the surface area of the sphere is $4\pi[R(t)\varpi]^2$.

Substituting this into Eq. (29.182), we find that

$$d_L = \varpi(1 + z), \quad (29.183)$$

where ϖ must be evaluated numerically using Eq. (29.168) and Eqs. (29.170–29.172). The luminosity distance, d_L , is the distance actually measured by the distance modulus $m - M$. Although the luminosity distance is *not* the same as either the present proper distance (Eqs. 29.148–29.150) or the coordinate distance (Eq. 29.3), the three distances do agree for $z \ll 1$.

Equation (29.176) shows that the luminosity distance is exactly given by

$$d_L(z) = \frac{c}{H_0}(1 + z)S(z). \quad (29.184)$$

Using our approximation to second order in z , Eq. (29.180),

$$d_L(z) \simeq \frac{cz}{H_0} \left[1 + \frac{1}{2}(1 - q_0)z \right] \quad (\text{for } z \ll 1). \quad (29.185)$$

Comparing this with Eq. (29.181), we see that the luminosity distance is approximately equal to the proper distance only when z is very small and the first term in each expansion dominates. For larger values of z , $d_p(z) < d_L(z)$. Ultimately, the exact expressions for both $d_p(z)$ and $d_L(z)$ are the most useful: Eqs. (29.169) and (29.184), respectively, evaluated by numerical integration.

The Redshift–Magnitude Relation

We are finally in a position to describe some of the most exciting observational tests of cosmology. The **redshift–magnitude relation** comes from using the luminosity distance with Eq. (3.6) for the distance modulus,

$$m - M = 5 \log_{10}(d_L/10 \text{ pc}) \quad (29.186)$$

Equation (29.184), along with $H_0 = 100h \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Eq. 29.13), quickly produces

$$\begin{aligned} m - M &= 5 \log_{10} \left[\frac{c}{(100 \text{ km s}^{-1} \text{ Mpc}^{-1})(10 \text{ pc})} \right] - 5 \log_{10}(h) \\ &\quad + 5 \log_{10}(1 + z) + 5 \log_{10}[S(z)] \\ &= 42.38 - 5 \log_{10}(h) + 5 \log_{10}(1 + z) + 5 \log_{10}[S(z)]. \end{aligned} \quad (29.187)$$

In the same manner, using the approximate Eq. (29.185) for the luminosity distance with Eq. (29.186) leads to, for $z \ll 1$,

$$\begin{aligned} m - M &\simeq 5 \log_{10} \left[\frac{c}{(100 \text{ km s}^{-1} \text{ Mpc}^{-1})(10 \text{ pc})} \right] - 5 \log_{10}(h) \\ &\quad + 5 \log_{10}(z) + 5 \log_{10} \left[1 + \frac{1}{2}(1 - q_0)z \right] \quad (\text{for } z \ll 1). \end{aligned}$$

Expanding the last term on the right in a Taylor series about $z = 0$ and keeping only the first-order terms in z results in

$$m - M \simeq 42.38 - 5 \log_{10}(h) + 5 \log_{10}(z) + 1.086(1 - q_0)z \quad (\text{for } z \ll 1). \quad (29.188)$$

Figure 29.26 shows the redshift z plotted on a logarithmic scale as a function of $m - M$. For $z \ll 1$, the redshift–magnitude relation is linear. Observations confirm the linearity of the $\log_{10}(z)$ term (which is just the Hubble law) for small z . Then, at larger z , the fourth term on the right-hand side of Eq. (29.188), $1.086(1 - q_0)z$, will cause the line to curve upward. Accurately measuring this departure from a straight line allows the value of the deceleration

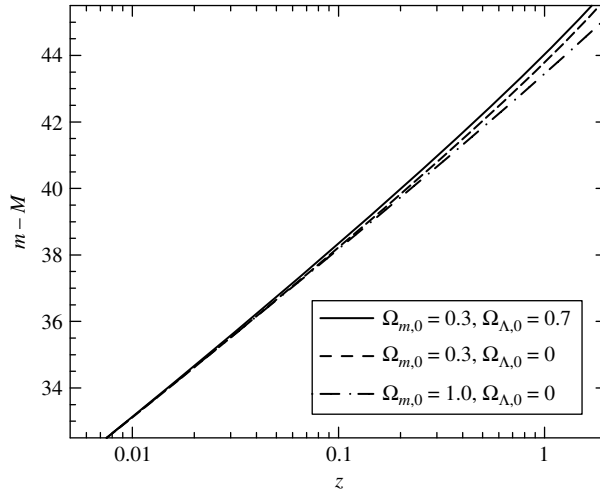


FIGURE 29.26 The redshift–magnitude relation for $h = 0.71$ and several values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$.

parameter to be determined. At still larger z , the curve is sensitive to the individual values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$.

The cosmological redshift affects the measurement of an object's spectrum because these observations are usually made within a specific wavelength region. For example, observations made in the V -band at 550 nm can be affected as the cosmological redshift brings shorter-wavelength radiation into the V band. This effect can be corrected for by adding a compensating term called the *K-correction* to Eq. (29.188) if the spectrum, I_λ , of the object is known.⁴²

In the mid-1990's two rival teams of astronomers—the Supernova Cosmology Project (SCP, led by Saul Perlmutter) and the High-Z Supernova Search Team (HZSNS Team, headed by Brian Schmidt)—made observations of Type Ia supernovae at cosmological distances. Both teams were astonished to discover that supernovae observed with a redshift of $z \approx 0.5$ were about 0.25 magnitude *dimmer* than expected for a universe with $\Omega_{m,0} \simeq 0.3$ and $\Lambda = 0$. The supernovae were farther away than they would be in this canonical decelerating universe. The possibility of an accelerating universe and a nonzero cosmological constant immediately leapt to their minds, but it took nearly a year of intense work to eliminate several plausible alternative explanations. For example, about 20% of the light from a distant supernova could be absorbed at high z by a hypothetical “gray dust.”⁴³ Or perhaps evolutionary effects were misleading the astronomers, since at high z we observe a younger generation of supernovae, formed in a younger galactic environment where heavy elements were less abundant. One by one, these possibilities were considered and eliminated. The redshift–magnitude diagram in Fig. 29.27 shows a more recent compilation of the results of these two teams. Both groups found that their analyses ruled out a flat universe with $\Omega_{m,0} = 1$ and $\Lambda = 0$ (the scenario championed by most theorists at the time) and were also incompatible with an open universe having $\Omega_{m,0} \simeq 0.3$ and $\Lambda = 0$. Instead, their findings favored a universe with $\Omega_{m,0} \simeq 0.3$ and $\Omega_{\Lambda,0} = 0.7$. Figure 29.28 shows, for each team, the location on the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane of the most likely set of values that are consistent with the high- z supernovae results. Their evidence for a nonzero cosmological constant is persuasive.

If we look at supernovae beyond $z_{\text{accel}} = 0.76$, when the universe started accelerating, we should find the signature of a decelerating universe. Figure 29.29 shows further results of observations of high- z supernovae, including six with $z > 1.25$. It includes SN 1997ff, the most distant supernova yet observed at $z = 1.7$. SN 1997ff and others in this high- z sample appear *brighter* than they would if the universe had expanded at a constant rate (i.e., with $\Omega = 0$), as expected for the deceleration phase of the early universe. These observations rule out the alternative explanations of acceleration, “gray dust” and evolutionary effects, as indicated in the figure. Clearly, astronomers will have to grapple with the implications of an accelerating universe whose dynamics are dominated by dark energy.

It is perhaps surprising that the values of the Hubble constant obtained from the redshift–magnitude diagrams ($H_0 \sim 70 \pm 10 \text{ km s}^{-1} \text{ Mpc}^{-1}$) are not better determined. The spread in the values of H_0 obtained by various groups using supernovae is due to their different calibrations of Cepheid distances. This systematic uncertainty does not affect the values

⁴²The *K*-correction was previously discussed in Section 25.2.

⁴³A “gray dust” of large grains would absorb light nearly equally at all wavelengths and so would produce no detectable reddening of a supernova's spectrum.

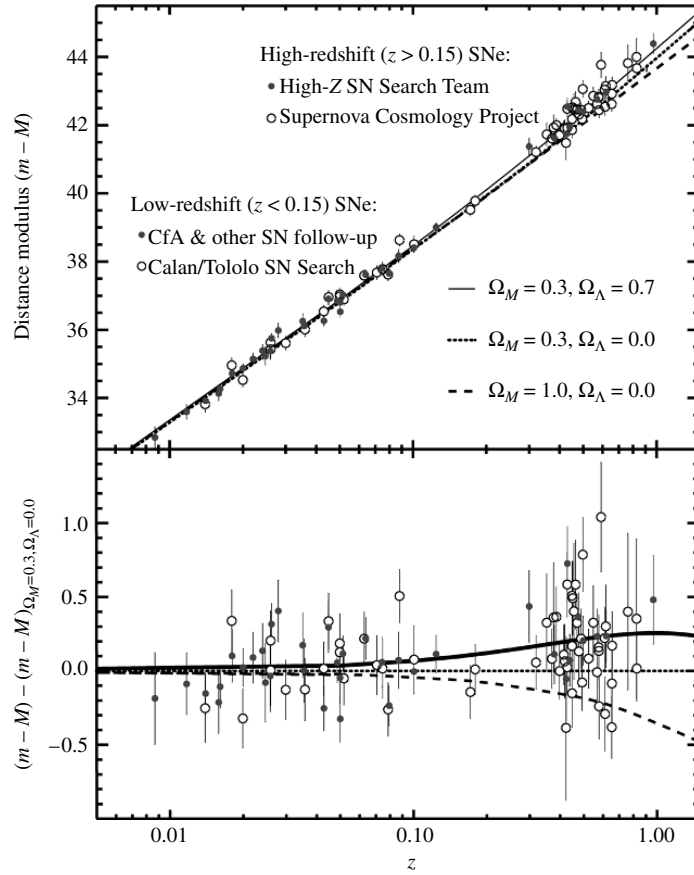


FIGURE 29.27 The redshift–magnitude relation measured for high- z supernovae. The K -correction has been applied to the apparent magnitudes. The lower graph shows the data after subtracting the theoretical curve for $\Omega_{m,0} = 0.3$, $\Omega_{\Lambda,0} = 0$. [Figure adapted from Perlmutter and Schmidt, *Supernovae and Gamma-Ray Bursters*, K. Weiler (ed.), *Lecture Notes in Physics*, 598, 195, 2003. Data from Perlmutter et al, *Ap. J.*, 517, 565, 1999 (SCP) and Riess et al, *A. J.*, 116, 1009, 1998 (HZSNS).]

of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ because these are determined by the *departures from linearity* in the redshift–magnitude diagram.

Angular Diameter Distance

Another measure of an object’s distance may be found by comparing its linear diameter D (assumed known) with its observed angular diameter θ (assumed small). The **angular diameter distance**, d_A is then defined to be

$$d_A \equiv \frac{D}{\theta}. \quad (29.189)$$

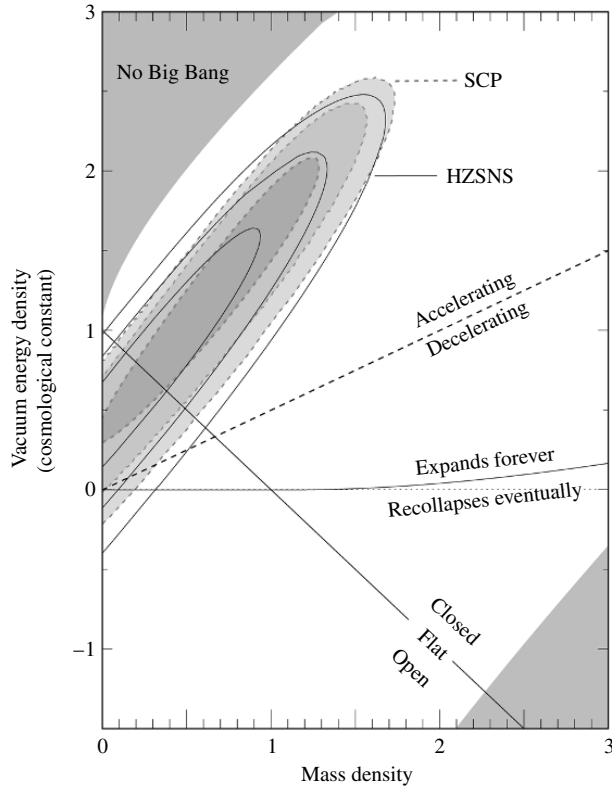


FIGURE 29.28 The location of the most probable values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ for high- z supernovae. The results of the SCP and the HZSNS Team are superimposed. [Figure adapted from Perlmutter and Schmidt, *Supernovae and Gamma-Ray Bursters*, K. Weiler (ed.), *Lecture Notes in Physics*, 598, 195, 2003. Data from Perlmutter et al, *Ap. J.*, 517, 565, 1999 (SCP) and Riess et al, *A. J.*, 116, 1009, 1998 (HZSNS).]

To place this into context with our previous results, consider a galaxy of redshift z located at comoving coordinate ϖ . We can use Eq. (17.18), $d\mathcal{L} = \sqrt{-(ds)^2}$, to find an expression for D , the proper distance from one side of the galaxy to the other. Integrating the Robertson–Walker metric (Eq. 29.106) across the galaxy in the plane of the sky with $dt = d\varpi = d\phi = 0$, we obtain

$$D = R(t_e)\varpi\theta = \frac{\varpi\theta}{1+z}.$$

[Note from Eq. (29.146) that for a flat universe this is the usual Euclidean geometric relation.] Of course, D is the galaxy’s diameter at the time t_e , when the light we observe was emitted. Since the light from the galaxy traveled a radial path to Earth, θ is the angular size of the galaxy as measured by astronomers. Equation (29.176) can be used to express the diameter

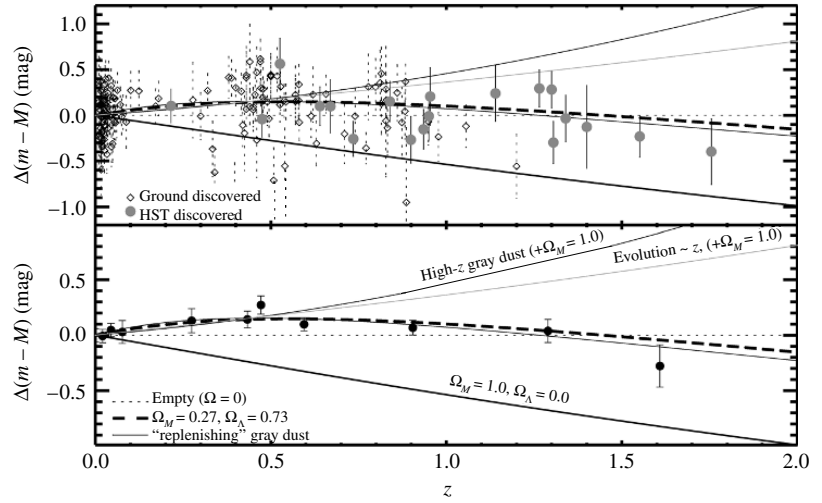


FIGURE 29.29 The redshift–magnitude relation measured for very high- z supernovae. The K -correction has been applied to the apparent magnitudes, and the theoretical curve for $\Omega = 0$ (a “coasting universe”) has been subtracted from the data. The lower graph illustrates the averages of binned data (grouped according to redshift) and compares them to curves of alternative models incorporating “gray dust” or evolutionary effects for supernovae. (Figure adapted from Riess et al, *Ap. J.*, 607, 665, 2004.)

as

$$D = \frac{c}{H_0} \frac{S(z)\theta}{1+z}. \quad (29.190)$$

Thus the angular diameter distance is

$$d_A = \frac{c}{H_0} \frac{S(z)}{1+z}. \quad (29.191)$$

From Eq. (29.183), we find that the angular diameter distance and the luminosity distance are related by

$$d_A = \frac{d_L}{(1+z)^2}. \quad (29.192)$$

Figure 29.30 shows a graph of θ in units of $H_0 D/c$,

$$\frac{c\theta}{H_0 D} = \frac{(1+z)}{S(z)}, \quad (29.193)$$

as a function of the redshift z for several model universes. It is surprising that the angular diameter of a galaxy does *not* continue to decrease with distance. In fact, beyond a certain redshift, the angular size actually *increases* with distance. This is due to the universe acting

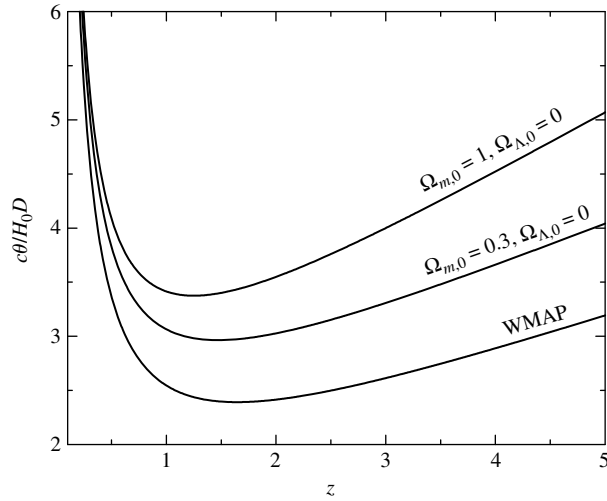


FIGURE 29.30 The angular diameter θ of a galaxy in units of $H_0 D/c$ for several values of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$.

as a sort of gravitational lens, enlarging the appearance of a galaxy beyond what would be expected in a static Euclidean universe. In principle, observations of galaxies of a known linear diameter D would allow observers to determine which values of cosmological parameters are preferred. In practice, however, galaxies do not have sharp boundaries, and they evolve as the universe ages. As of this writing, the most productive use of the angular diameter distance has been coupled with observations of the Sunyaev–Zel’dovich effect.

The Sunyaev–Zel’dovich effect provides an independent determination of the Hubble constant. Measurements of $\Delta T/T_0$ (Eq. 29.64) along with the X-ray flux F_X and temperature T_e of the intracluster gas in rich clusters of galaxies can be used to model the physical properties of the cluster. Comparing the calculated diameter D of the cluster with its measured angular diameter θ yields d_A , the cluster’s angular diameter distance. On the other hand, the measured X-ray flux from the cluster and Eq. (27.20) for the X-ray luminosity of the intracluster gas determine the cluster’s luminosity distance. The connection between these two distances, Eq. (29.192), can then be used to calculate the Hubble constant. In Problem 29.55 you will derive

$$H_0 = Cf(z) \frac{F_X T_e^{3/2}}{\theta (\Delta T/T_0)^2},$$

where $f(z)$ is a function of the redshift z of the cluster and C is a constant. One team of astronomers measured⁴⁴ the value of the Hubble constant for five clusters and obtained an average of $H_0 = 65 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with the values obtained from other recent measurements.

⁴⁴See Jones et al. (2005) for details of how H_0 was determined.

SUGGESTED READING

General

- Alpher, R. A., and Herman, R. C., "Evolution of the Universe," *Nature*, 162, 774, 1948.
- Goldsmith, Donald, *The Runaway Universe*, Perseus Publishing, Cambridge, MA, 2000.
- Harrison, Edward, *Cosmology*, Second Edition, Cambridge University Press, Cambridge, 2000.
- Harrison, Edward, *Darkness at Night*, Harvard University Press, Cambridge, MA, 1987.
- Kirshner, Robert, *The Extravagant Universe*, Princeton University Press, Princeton, NJ, 2002.
- Krauss, Lawrence M., "Cosmological Antigravity," *Scientific American*, January 1999.
- Lineweaver, Charles, and Davis, Tamara, "Misconceptions about the Big Bang," *Scientific American*, March 2005.
- Silk, Joseph, *The Big Bang*, Third Edition, W. H. Freeman and Company, New York, 2001.
- Silk, Joseph, *On the Shores of the Unknown*, Cambridge University Press, Cambridge, 2005.
- Weinberg, Steven, *The First Three Minutes*, Second Edition, Basic Books, New York, 1988.

Technical

- Bennett, C. L., "First-Year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Preliminary Maps and Basic Results," *The Astrophysical Journal Supplement*, 148, 1, 2003.
- Binney, James, and Tremaine, Scott, *Galactic Dynamics*, Princeton University Press, Princeton, NJ, 1987.
- Carlstrom, John E., Holder, Gilbert P., and Reese, Erik D., "Cosmology with the Sunyaev-Zel'dovich Effect," *Annual Review of Astronomy and Astrophysics*, 40, 643, 2002.
- Davis, Tamara M., Lineweaver, Charles H., and Webb, John K., "Solutions to the Tethered Galaxy Problem in an Expanding Universe and the Observation of Receding Blueshifted Objects," *American Journal of Physics*, 71, 358, 2003.
- Dicke, R. H., Peebles, P. J. E., Roll, P. G., and Wilkinson, D. T., "Cosmic Black-body Radiation," *The Astrophysical Journal*, 142, 414, 1965.
- Felten, James E., and Isaacman, Richard, "Scale factors $R(t)$ and Critical Values of the Cosmological Constant Lambda in Friedmann Universes," *Reviews of Modern Physics*, 58, 689, 1986.
- Foley, Ryan J., et al., "A Definitive Measurement of Time Dilation in the Spectral Evolution of the Moderate-Redshift Type Ia Supernova 1997ex," *The Astrophysical Journal Letters*, 626, L11, 2005.
- Garnavich, Peter M., et al., "Supernova Limits on the Cosmic Equation of State," *The Astrophysical Journal*, 509, 74, 1998.
- Gott, J. Richard III, et al., "A Map of the Universe," *The Astrophysical Journal*, 624, 463, 2005.
- Harrison, Edward R., "Why Is the Sky Dark at Night?" *Physics Today*, February 1974.

- Jones, Michael E., et al., “ H_0 from an Orientation-Unbiased Sample of SZ and X-ray Clusters,” *Monthly Notices of the Royal Astronomical Society*, 357, 518, 2005.
- Kolb, Edward W., and Turner, Michael S., *The Early Universe*, Westview Press, 1994.
- Loeb, Abraham, “Long-Term Future of Extragalactic Astronomy,” *Physical Review D*, 65, 047301, 2002.
- Peacock, John A., *Cosmological Physics*, Cambridge University Press, Cambridge, 1999.
- Peebles, P. J. E., *Principles of Physical Cosmology*, Princeton University Press, Princeton, NJ, 1993.
- Penzias, A. A., and Wilson, R. W., “A Measurement of Excess Antenna Temperature at 4080 Mc/s,” *The Astrophysical Journal*, 142, 419, 1965.
- Perlmutter, S., et al., “Measurements of Ω and Λ from 42 High-Redshift Supernovae,” *The Astrophysical Journal*, 517, 565, 1999.
- Perlmutter, S., and Schmidt, B. P., “Measuring Cosmology with Supernovae,” *Supernovae and Gamma-Ray Bursters*, K. Weiler (ed.), *Lecture Notes in Physics*, 598, 195, 2003.
- Raine, D. J., and Thomas, E. G., *An Introduction to the Science of Cosmology*, Institute of Physics Publishing, Philadelphia, 2001.
- Riess, Adam G., et al., “Type Ia Supernova Discoveries at $z > 1$ from the *Hubble Space Telescope*: Evidence for Past Deceleration and Constraints on Dark Energy Evolution,” *The Astrophysical Journal*, 607, 665, 2004.
- Riess, Adam G., et al., “Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant,” *The Astronomical Journal*, 116, 1009, 1998.
- Ryden, Barbara, *Introduction to Cosmology*, Addison-Wesley, New York, 2003.
- Spinrad, Hyron, Dey, Arjun, and Graham, James R., “Keck Observations of the Most Distant Galaxy: 8C 1435+63 at $z = 4.25$,” *The Astrophysical Journal Letters*, 438, L51, 1995.

PROBLEMS

- 29.1** It might be argued that the inverse square law for light, Eq. (3.2), would provide a solution to Olbers’s paradox. To see that this is not so, consider a uniform distribution of stars with n stars per unit volume, each of luminosity L . Imagine that two thin, spherical shells of stars with radii r_1 and r_2 are centered on Earth; let the thickness of each shell be Δr . Show that the same energy flux reaches Earth from each shell.
- 29.2** Suppose that all of the matter in the universe were converted into energy in the form of blackbody radiation. Take the average density of matter to be the WMAP value for the density of baryonic matter, $\rho_{b,0}$. Use Eq. (9.7) for the energy density of blackbody radiation to find the temperature of the universe in this situation. At what wavelength would the blackbody spectrum peak? In what region of the electromagnetic spectrum is this wavelength found? Explain how your result may be applied to Olbers’s paradox.
- 29.3** Show by substitution that Eqs. (29.32) and (29.34) are solutions to Eq. (29.11) for a closed universe ($k > 0$).
- 29.4** Show by substitution that Eqs. (29.36) and (29.38) are solutions to Eq. (29.11) for an open universe ($k < 0$).

- 29.5** Derive Eqs. (29.33) and (29.35) from Eqs. (29.32) and (29.34), respectively.
- 29.6** Derive Eqs. (29.37) and (29.39) from Eqs. (29.36) and (29.38), respectively.
- 29.7** (a) Use Eq. (29.11) to find an expression for the maximum scale factor R in a closed universe. Does your answer agree with Eq. (29.33)?
- (b) Find the lifetime of a closed universe (expressed as a multiple of the Hubble time, t_H) as a function of the density parameter, Ω_0 .
- 29.8** Derive Eqs. (29.40–29.42) for the age of the universe using Eqs. (29.31), (29.33), (29.35), (29.37), (29.39), and (29.4).
- 29.9** Consider a one-component universe of pressureless dust.
- (a) Show that

$$\Omega(t) = \frac{\rho(t)}{\rho_c(t)} = 1 + \frac{kc^2}{(dR/dt)^2}, \quad (29.194)$$

which describes how Ω varies with time. What does this have to say about the nature of the early universe?

- (b) Show that $dR/dt \rightarrow \infty$ as $t \rightarrow 0$. What does this say about the difference between a closed, a flat, and an open universe at very early times?
- 29.10** For a one-component universe of pressureless dust, show that

$$\frac{1}{\Omega} - 1 = \left(\frac{1}{\Omega_0} - 1 \right) (1+z)^{-1}. \quad (29.195)$$

What happens as z increases?

- 29.11** Show that in the limit $(1+z) \gg 1/\Omega_0$, Eq. (29.41) reduces to Eq. (29.43). *Hint:* First write Eq. (29.41) in terms of a variable $u \equiv 1/[\Omega_0(1+z)]$, and then expand the equation in a Taylor series about $u = 0$. You may find

$$\cos^{-1}(1-x) = \sqrt{2}x^{1/2} + \frac{\sqrt{2}}{12}x^{3/2} + \dots$$

for $x \ll 1$ to be useful.

- 29.12** Derive the acceleration equation, Eq. (29.51).
- 29.13** Assuming that $P = 0$, show that Eq. (29.51) for the acceleration of a mass shell can be found from Newton's second law by considering the gravitational force on an expanding shell.
- 29.14** Consider a model of the universe consisting of neutral hydrogen atoms for which the average (rms) speed of the atoms is 600 km s^{-1} (approximately the speed of the Local Group relative to the Hubble Flow). Show that $\rho \gg P/c^2$ for the gas. For an adiabatically expanding universe, for what value of R and z will $\rho = P/c^2$?
- 29.15** By inserting the equation of state $P = w\rho c^2$ into the fluid equation, Eq. (29.50), show that $R^{3(1+w)}\rho = \text{constant} = \rho_0$, where ρ_0 is the present value of ρ .
- 29.16** Show that for a pressureless dust universe, $q(t) = \frac{1}{2} \Omega(t)$, which is Eq. (29.55).
- 29.17** The deuterium (${}^2\text{H}$) nucleus is not very tightly bound.

- (a) Calculate the binding energy of the deuterium nucleus, using values of $m_H = 1.007825$ u, $m_n = 1.008665$ u, and $m_D = 2.014102$ u.
- (b) What is the wavelength of a photon with this energy?
- (c) From Wien's law, at what temperature is this the characteristic energy of a blackbody photon?
- 29.18** The carbon absorption lines that are formed when the light from distant quasar Q1331+170 passes through an intergalactic cloud have been measured by Antoinette Songaila and her colleagues. The relative strengths of the lines indicate that the temperature of the cloud is 7.4 ± 0.8 K, and the lines show a redshift of $z = 1.776$. How does the temperature of the cloud compare with the temperature of the CMB at that redshift? (If there are sources of heating for the cloud in addition to the CMB, then its temperature must be considered as an *upper limit* to the temperature of the CMB.)
- 29.19** In 1941, microwave observations detected absorption lines due to cyanogen molecules (CN) in molecular clouds. A cyanogen molecule has three first excited rotational states, each of which is degenerate and has an energy that is 4.8×10^{-4} eV above the ground state. An analysis of the absorption lines shows that for every 100 molecules in the ground state, there are 27 others that are in one of the three first excited states. Assuming that the molecular clouds are in thermal equilibrium with the CMB, use the Boltzmann equation (Eq. 8.6) to estimate the temperature of the CMB.
- 29.20** Channel 6 on your television consists of radio waves with wavelengths between 3.41 m and 3.66 m. Consider a 25,000-watt television station located 70 km from your home. Use Eq. (9.5) for the energy density of blackbody radiation to estimate the ratio of the number of channel 6 photons to the number of CMB photons that your television antenna picks up in this wavelength band. (*Hint:* For the television broadcast, recall that the energy density of an electromagnetic wave is related to the time-averaged Poynting vector by $u = \langle S \rangle / c$.)
- 29.21** Use Eq. (4.32) for the relativistic Doppler shift to derive Eq. (29.61). Show that Eq. (29.61) reduces to Eq. (29.62) when $v \ll c$.
- 29.22** Calculate the magnitude of the variation in the temperature of the CMB due to the Sun's peculiar velocity.
- 29.23** In this problem you will approximate the physics that produces the Sunyaev–Zel'dovich effect.
- (a) First, estimate the shift in the frequency of a low-energy CMB photon as it is scattered by a high-energy electron (inverse Compton scattering) of speed v_e in the hot intracluster gas of a rich cluster of galaxies. Although photons can be scattered from any direction into any direction, we will consider the four equally likely situations shown in Fig. 29.31. Show that the *average* shift in the frequency of these four photons is

$$\frac{\Delta \nu}{\nu} = \frac{v_e^2}{c^2} = 3 \frac{kT_e}{m_e c^2},$$

where T_e is the temperature of the electron gas. Evaluate this expression using $T_e = 10^8$ K, and explain why you could assume that in the rest frame of the electron, the change in the wavelength ($\sim \lambda_C$) of the photon could be neglected.

- (b) About what fraction of the CMB photons will be scattered as they pass through the intracluster gas? Assume an electron number density of $n_e = 10^4 \text{ m}^{-3}$ and a cluster radius of 3 Mpc.

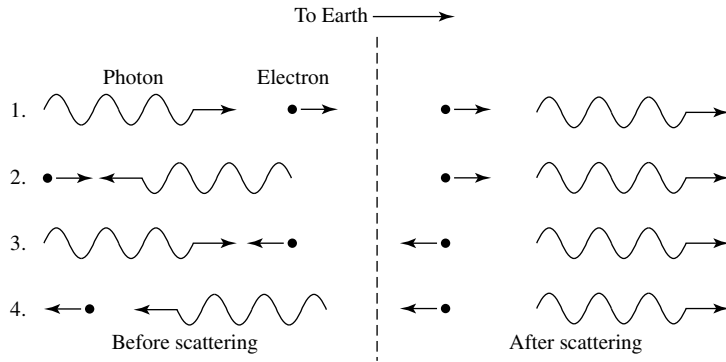


FIGURE 29.31 Inverse Compton scattering of a CMB photon by a high-energy electron.

- (c) Use the increase $\Delta\nu$ of the peak frequency with Wien's law (Eq. 29.59) to obtain an approximate expression for the effective decrease in the temperature of the CMB, $\Delta T/T_0$ (the Sunyaev–Zel'dovich effect).
- 29.24** Show that in the general equation of state $P = wu$ (Eq. 29.52), $w = 1/3$ for relativistic particles ($E \gg mc^2$). *Hint:* The pressure integral, Eq. (10.8), may prove useful.
- 29.25** Consider a comoving sphere whose surface expands with the universe. Let it be centered at the origin and filled with CMB photons. Show that Eq. (29.81), $R^4 \rho_{\text{rel}} = \rho_{\text{rel},0}$, is consistent with the conservation of energy within the sphere.
- 29.26** Some quantities obey an exponential time-behavior of the form $f(t) = f_0 e^{t/\tau}$, where τ is the *characteristic time* for the system under consideration.
- (a) Show that

$$\tau = \left(\frac{1}{f} \frac{df}{dt} \right)^{-1}.$$

This expression can be used to define a characteristic time for any function, regardless of whether its behavior is exponential. (See the analogous discussion of the pressure scale height, H_p , in Section 10.4.)

- (b) Use the scale factor, $R(t)$, to show that the characteristic time for the expansion of the universe is $\tau_{\text{exp}}(t) = 1/H(t)$.
- (c) Assuming a flat universe containing only matter and radiation, find an expression (valid in both the radiation era and the matter era) for the characteristic expansion time τ_{exp} as a function of the scale factor R .
- 29.27** (a) Show that deep in the radiation era when $R \ll R_{r,m}$, Eq. (29.84) is well approximated by Eq. (29.86).
- (b) Solve the Friedmann equation for a flat, one-component universe that contains only relativistic particles, and compare your result with Eq. (29.86).
- 29.28** Use a procedure similar to that used to obtain Eq. (29.28) to show that a one-component universe of relativistic particles is flat in the limit $z \rightarrow \infty$.

- 29.29** Assuming that the present density of baryonic matter is given by Eq. (29.17), what was the density of matter at the time of Big Bang nucleosynthesis, when $T \sim 10^{10}$ K?
- 29.30** One factor that contributed to the cessation of the reactions that formed neutrons at roughly 10^{10} K [Eqs. (29.93–29.95)] was the annihilation of electron–positron pairs that occurred at that time. When the temperature became too low, the electron–positron pairs could not be replaced by pair production. (This removed the supply of electrons that could combine with protons to form neutrons.) By setting the characteristic thermal energy of a photon, kT , equal to the rest energy of an electron–positron pair, estimate the temperature below which an annihilated pair will not readily be replaced.
- 29.31** In this problem, you will show that when the temperature of the universe was about 10^9 K, all of the neutrons would have combined with protons to form helium nuclei.
- (a) Using arguments similar to those leading up to Eq. (9.12), show that the number of collisions between a neutron and a proton that occur within a time Δt is $n_p \sigma v \Delta t$, where n_p is the number density of protons, σ is the neutron’s collision cross section, and v is the speed of the neutron.
- (b) Evaluate $n_p \sigma v \Delta t$. If the result is $\gg 1$, then each neutron had ample opportunity to combine with a proton. Let Δt be the characteristic timescale of the universe at the time of helium formation, and use $\sigma = \pi(2r)^2$, where $r \simeq 10^{-15}$ m is the radius of a neutron. The number density of protons can be estimated from the baryonic mass density when $T = 10^9$ K.
- 29.32** (a) Use the cross section for electron scattering (Eq. 9.20) to find an expression for the average time between scatterings of a photon by free electrons.
- (b) Assuming that the electrons remain free, at what value of R and z will the average time between scatterings equal the characteristic expansion time τ_{exp} of Problem 29.26? (Use WMAP values.) What is the age of the universe when this occurs? This is when a flat universe of matter and radiation would have become transparent due solely to its expansion (no recombination).
- 29.33** Solve Eq. (29.101) for a composition of pure hydrogen to find the temperature when half of the electrons and protons have combined to form neutral atoms.
- 29.34** Calculate the time of decoupling, t_{dec} , for a universe of matter and radiation using the WMAP values for $z_{\text{dec}} = 1089$ and other quantities. Compare your answer with the WMAP result of 379^{+8}_{-7} kyr.
- 29.35** Using WMAP values for a universe of matter and radiation, estimate the time interval Δt between when recombination began (say, when 99% of the hydrogen atoms were ionized) and when recombination ended (say, when 1% of the hydrogen atoms were ionized). What is the difference Δz between the values of the redshift z at these two times? This is the thickness (in terms of z) of the “surface” of last scattering. Compare your answers with the WMAP results of $\Delta t = 118^{+3}_{-2}$ kyr and $\Delta z = 195 \pm 2$. Assume a composition of pure hydrogen.
- 29.36** Suppose that Earth were a perfectly smooth sphere. If you drew a circle of radius $D = 100$ meters on Earth’s surface, what discrepancy would you find between the expected and measured values of the circle’s circumference?
- 29.37** Follow a procedure similar to that used to obtain Eq. (29.28) to derive an expression for the total density parameter, $\Omega(z)$, as a function of z . Verify that your expression reduces to

Eq. (29.28) if $\Omega_{\text{rel},0} = \Omega_{\Lambda,0} = 0$. What does your expression say about the geometry of an early, three-component universe?

- 29.38** Use the Robertson–Walker metric, Eq. (29.106), to show that the proper area ($dt = 0$) of a spherical surface, centered at the origin and passing through comoving coordinate ϖ , is $4\pi[R(t)\varpi]^2$.
- 29.39** Einstein originally introduced the cosmological constant Λ to stabilize his model of a pressureless dust universe against expansion or contraction.
- (a) Find an expression for Λ in terms of the density ρ_m of a static model of a pressureless dust universe with a cosmological constant.
 - (b) Find an expression for the curvature k for this static model. Is this model universe closed, open, or flat?
 - (c) Explain why Einstein’s static model is in an unstable equilibrium, so any departure from equilibrium (expansion or contraction) will tend to increase.
- 29.40** Evaluate Ω_m , Ω_{rel} , and Ω_{Λ} at the time of decoupling ($z = 1089$) using WMAP values.
- 29.41** Show that Eq. (29.122) may be written as

$$H = H_0 \left[\sum_i \Omega_{i,0} (1+z)^{3(1+w_i)} + (1 - \Omega_0) (1+z)^2 \right]^{1/2}, \quad (29.196)$$

where w is the coefficient from the equation of state $P_i = w_i \rho_i c^2$ and the “ i ” subscripts identify one of the components of the universe (i.e., pressureless dust, relativistic particles, or dark energy).

- 29.42** Derive Eq. (29.123) for a general expression of the deceleration parameter,

$$q(t) = \frac{1}{2} \sum_i (1 + 3w_i) \Omega_i(t).$$

- 29.43** Use the acceleration equation to show that the acceleration of the universe changed sign (from negative to positive) when the scale factor was

$$R_{\text{accel}} = \left(\frac{\Omega_{m,0}}{2\Omega_{\Lambda,0}} \right)^{1/3}.$$

Evaluate the value of R_{accel} and z_{accel} at this time with WMAP values.

- 29.44** (a) Use Eq. (29.129) to find an expression for the lookback time, t_L , as a function of the redshift z .
- (b) Figure 28.17 shows the comoving space density of active galactic nuclei (AGN) as a function of redshift. Using your expression for the lookback time with WMAP values, replot the “ChoMP + CDF + ROSAT” data (marked with filled circles) with t_L/t_H on the horizontal axis (the lookback time as a fraction of the Hubble time). How would you characterize the decline in the space density of AGN with increasing lookback time?
- 29.45** The cosmological constant becomes dominant as the scale factor R becomes increasingly larger in the Λ era.
- (a) Show that the Hubble parameter is a constant in a flat universe deep in the Λ era.

- (b) Suppose that, starting today ($t = t_0$, when $R = 1$), only the cosmological constant contributes to the Friedmann equation.⁴⁵ Solve the Friedmann equation and show that for $\Lambda > 0$, the scale factor will increase exponentially.
- (c) Use WMAP values to evaluate the characteristic time for the exponential expansion (cf. Problem 29.26).

29.46 In the matter era, the distance to the particle horizon for a closed, one-component universe of pressureless dust is given by

$$d_h(z) = \frac{c}{H_0(1+z)\sqrt{\Omega_0-1}} \cos^{-1} \left[1 - \frac{2(\Omega_0-1)}{\Omega_0(1+z)} \right]. \quad (29.197)$$

In this problem, you will derive this expression for d_h . First change variables in Eq. (29.153) to obtain

$$d_h(t) = R(t) \int_0^{\frac{1}{1+z}} \frac{c dR}{R(dR/dt)},$$

where the limits of integration range from $R = 0$ (at $t = 0$) to $R = 1/(1+z)$ (at time t). Then show that

$$\left(\frac{dR}{dt} \right)^2 = H_0^2 \left(\frac{\Omega_0}{R} - \Omega_0 + 1 \right),$$

and make this substitution into the denominator of the integral. You may find

$$\int \frac{dx}{\sqrt{bx - ax^2}} = \frac{1}{\sqrt{a}} \left[\cos^{-1} \left(1 - \frac{2ax}{b} \right) - \frac{\pi}{2} \right]$$

to be useful.

- 29.47** Use the results of Problem 29.46 to find the ratio of the distance to the particle horizon to the circumference of a closed, matter-dominated universe. What happens at very early times, as $z \rightarrow \infty$? Show that at the time of maximum expansion (just before the closed universe begins to collapse), this ratio is equal to one-half. This means that at the moment of the Big Crunch ending the collapse, the particle horizon encompasses the entire universe.
- 29.48** Using Eq. (29.163) for the comoving coordinate, ϖ , of a photon just now arriving from the present particle horizon in a flat universe, find the maximum proper distance of the photon during its journey. Express your answer as a fraction of the model's particle horizon, $d_{h,0} = 3ct_0$. At what time (t/t_0) is the photon at this distance? Carefully explain the meaning of the phrase “just now arriving from the present particle horizon.”
- 29.49** Consider the (unrealistic) model of a flat, one-component universe of pressureless dust, as described in Section 29.1.

- (a) Show that for this model,

$$\varpi = \frac{2c}{H_0} \left(1 - \frac{1}{\sqrt{1+z}} \right). \quad (29.198)$$

- (b) Find an expression for the proper distance to an object with redshift z for this model.

⁴⁵Such a universe is called a *de Sitter universe*, named for the Dutch mathematician Willem de Sitter (1872–1934).

- (c) Find an expression for the horizon distance in this model. Evaluate this using WMAP values, and compare your result with the more accurate value of Eq. (29.159).

29.50 Derive Mattig's relation for a one-component universe of pressureless dust,

$$\varpi = \frac{2c}{H_0} \frac{1}{1+z} \frac{1}{\Omega_0^2} \left[\Omega_0 z - (2 - \Omega_0)(\sqrt{1 + \Omega_0 z} - 1) \right], \quad (29.199)$$

or in terms of the deceleration parameter $q_0 = \Omega_0/2$ (valid for this one-component model),

$$\varpi = \frac{c}{H_0} \frac{1}{1+z} \frac{1}{q_0^2} \left[q_0 z - (1 - q_0)(\sqrt{1 + 2q_0 z} - 1) \right]. \quad (29.200)$$

Show that this is valid for a flat, an open, and a closed one-component universe of pressureless dust. Note that Eq. (29.198) in the previous problem and Eq. (29.199) are in agreement when $\Omega_0 = 1$.

29.51 Show that Eq. (29.178) is the Taylor-series expansion of Eq. (29.168) about $z = 0$.

29.52 In this problem, you will carry out a general derivation of Eq. (29.180) without reference to any specific model of the universe.

- (a) Expand the scale factor, $R(t)$, in a general Taylor series about the present time, t_0 , and obtain

$$\begin{aligned} R(t) &= R(t_0) + \left. \frac{dR}{dt} \right|_{t_0} (t - t_0) + \frac{1}{2} \left. \frac{d^2 R}{dt^2} \right|_{t_0} (t - t_0)^2 + \cdots \\ &= 1 - H_0(t_0 - t) - \frac{1}{2} H_0^2 q_0 (t_0 - t)^2 + \cdots \end{aligned} \quad (29.201)$$

- (b) Show that $1/R(t)$ is then given by

$$\frac{1}{R(t)} = 1 + H_0(t_0 - t) + H_0^2 \left(1 + \frac{1}{2} q_0 \right) (t_0 - t)^2 + \cdots$$

[Hint: Use long division to divide 1 by $R(t)$.]

- (c) Use Eq. (29.142), $1/R(t) = 1 + z$, with the result of part (b) to write the expansion for z about the present time. Now solve for $t_0 - t$ and express the result as a series in z to get

$$t_0 - t = \frac{z}{H_0} - \left(1 + \frac{1}{2} q_0 \right) \frac{z^2}{H_0} + \cdots$$

- (d) Consider a photon that is emitted at comoving coordinate ϖ at time t and received on Earth at the present time t_0 . Use an equation similar to Eq. (29.138) to find an approximate expression for ϖ . You need only use the first two terms in the expansion in part (b) for $1/R(t)$ in the left-hand integral. For the right-hand integral, use a two-term Taylor series for $1/\sqrt{1 - k\varpi^2}$. You should find that

$$\varpi = c(t_0 - t) \left[1 + \frac{1}{2} H_0(t_0 - t) \right] + \cdots$$

- (e) By substituting the expression for $t_0 - t$ from part (c) into this equation for ϖ , show that, to second order in z ,

$$\varpi = \frac{cz}{H_0} \left[1 - \frac{1}{2} (1 + q_0) z \right] + \cdots$$

This result is very important because it does not rely on any particular model of the universe. It is valid even if the cosmological constant, Λ , is not equal to zero.

29.53 Use Eq. (29.196) from Problem 29.41 to show that the luminosity distance may be written as

$$d_L = \frac{c(1+z)}{H_0\sqrt{|1-\Omega_0|}} \text{sinn} \left\{ \sqrt{|1-\Omega_0|} \int_0^z \frac{dz'}{[\sum_i \Omega_{i,0}(1+z')^{3(1+w_i)} + (1-\Omega_0)(1+z')^2]^{1/2}} \right\}, \quad (29.202)$$

where

$$\begin{aligned} \text{sinn}(x) &\equiv \sinh(x) && \text{if } \Omega_0 < 1, \\ &\equiv x && \text{if } \Omega_0 = 1, \\ &\equiv \sin(x) && \text{if } \Omega_0 > 1. \end{aligned}$$

This is Eq. (1) of Garnavich et al. (1998), which explains how this equation may be used with the distance modulus, Eq. (29.186), to place limits on the value of w_Λ for the equation of state of dark energy. The WMAP result is that $w_\Lambda < -0.78$.

29.54 Assume a flat, one-component universe of pressureless dust for this problem.

(a) Show that the angular diameter observed for an extended object of linear diameter D at redshift z is

$$\theta = \frac{H_0 D}{2c} \frac{(1+z)^{3/2}}{\sqrt{1+z}-1}.$$

(b) Find the value of the redshift for which θ is a minimum. Compare your result with Fig. 29.30.

(c) What is the smallest value of θ when observing a cluster of galaxies with a diameter of 1 Mpc? Use $h = 0.71$.

29.55 Model the hot intragalactic gas in a rich cluster of galaxies as a homogeneous sphere of radius R and temperature T_e . Let F_X be the X-ray flux observed for the gas, and let θ be the angular diameter of the gas, as observed from Earth.

(a) Show that if the Sunyaev–Zel’dovich effect, $\Delta T/T_0$, is measured for the cluster, then the Hubble constant may be calculated as

$$H_0 = Cf(z) \frac{F_X T_e^{3/2}}{\theta (\Delta T/T_0)^2}.$$

where $f(z)$ is a function of the redshift z of the cluster and C is a constant factor you must determine.

(b) Use the data in Fig. 29.12 for the clusters, together with $\theta = 46''$, $F_X = 5.77 \cdot 10^{-15} \text{ W m}^{-2}$, and $kT_e = 8.0 \text{ keV}$ for Abell 697, and $\theta = 69''$, $F_X = 7.16 \cdot 10^{-15} \text{ W m}^{-2}$, and $kT_e = 7.2 \text{ keV}$ for Abell 2218 (from Jones et al., 2005) to evaluate h , the Hubble parameter. How do your answers compare with the WMAP value of $h = 0.71$?

COMPUTER PROBLEMS

- 29.57** In Eq. (29.166), the limits of integration on the left-hand side span a time interval that lies entirely in the past. Use Eq. (29.132) to approximate the scale factor on the left-hand side. On the right-hand side, consider the case of $t_{\text{mva}} > t_0$, so the limits of integration span a time interval that lies entirely in the future. Use Eq. (29.133) for the scale factor on the right-hand side. [Note that very distant objects have already emitted the last photon that will ever reach us ($t_{\text{mva}} < t_0$), so we must restrict our attention to nearer objects to ensure that $t_{\text{mva}} > t_0$.] Show that

$$t_{\text{mva}} \simeq \frac{t_H}{\sqrt{\Omega_{\Lambda,0}}} \ln \left[\left(\frac{\sqrt{\Omega_{m,0}}}{2\sqrt{\Omega_{\Lambda,0}}} \right)^{1/3} \left(\frac{\sqrt{1+z}}{\sqrt{1+z-1}} \right) \right].$$

Using WMAP values, what is the largest redshift for which $t_{\text{mva}} > t_H$? Find the maximum visible age, in units of t_H , for sources at values of z of 0.1, 0.5, 1, and 1.5.

- 29.58** Use WMAP values for this problem.

- (a) On a single graph, plot the luminosity distance, the present proper distance, the angular diameter distance, the relativistic Hubble law distance estimate (Eq. 27.7), and the non-relativistic Hubble law distance (Eq. 27.8) for values of z between 0 and 4. Express these distances in units of c/H_0 .
- (b) From your results, determine the value of z when the relativistic Hubble law distance estimate differs from the proper distance by more than 10%.

- 29.59** A distant radio galaxy, 8C 1435+63, has a redshift of $z = 4.25$. Assume WMAP values for this problem.

- (a) How old was the universe at this redshift? Express your answer both in terms of years and as a fraction of the present age of the universe.
- (b) What is the present proper distance (in Mpc) to 8C 1435+63?
- (c) What was the proper distance (in Mpc) to 8C 1435+63 when its light was emitted?
- (d) What is the luminosity distance to 8C 1435+63?
- (e) What is the angular diameter distance to 8C 1435+63?
- (f) The angular diameter of the nucleus of 8C 1435+63 is about $5''$. What is the linear diameter of the galaxy (in units of kpc)?
- (g) Suppose the galaxy's redshift were $z = 1$. What would its linear diameter be [using the same angular diameter as in part (d)]?

Further information about 8C 1435+63, which may be the progenitor of a cD elliptical galaxy, can be found in Spinrad, Dey, and Graham (1995).

CHAPTER

30

The Early Universe

30.1 *The Very Early Universe and Inflation*

30.2 *The Origin of Structure*

30.1 ■ THE VERY EARLY UNIVERSE AND INFLATION

In Chapter 29 we developed the mathematical machinery that describes the expansion of an isotropic, homogeneous universe consisting of nonrelativistic particles (baryonic and dark), relativistic particles (photons and neutrinos), and dark energy. This chapter deals with the emergence of structure from the featureless cauldron of the Big Bang.

Much happened during the universe's first fraction of a second, and the nature of that earliest environment is still only partially understood. The ideas of modern particle physics that describe these early epochs have also made successful predictions about the types and numbers of elementary particles that exist today. It is to these theories that we turn (with some measure of confidence) in hopes of glimpsing the engines of creation.

Fundamental Particles

According to the **Standard Model** of particle physics, there are three kinds of fundamental (not composite) particles.

- The **leptons** are the charged leptons $e^\pm$, $\mu^\pm$, and $\tau^\pm$, and the neutrinos ν_e , ν_μ , and ν_τ , together with their antineutrinos $\bar{\nu}_e$, $\bar{\nu}_\mu$, and $\bar{\nu}_\tau$. Leptons are fermions.
- There are six **quarks**: up, down, strange, charm, bottom, and top, along with their antiquarks. Each quark comes in three “colors” (three choices of an internal degree of freedom). Particles made of quarks are called **hadrons**. There are two types of hadrons: **baryons** (made of three quarks) and **mesons** (formed by a quark–antiquark pair). Baryons are fermions, while mesons are bosons.
- The **force-carrying particles** consist of the photon, eight different gluons (particles that mediate the strong interaction and bind quarks together), three vector gauge bosons ($W^\pm$ and Z^0) that mediate the weak interaction, and the scalar Higgs boson (which has yet to be confirmed by experiment). All of these particles are bosons.

Table 30.1 summarizes some of the properties of these particles. Current technology can reproduce the temperatures, energies, and densities that prevailed back to the **quark–hadron**